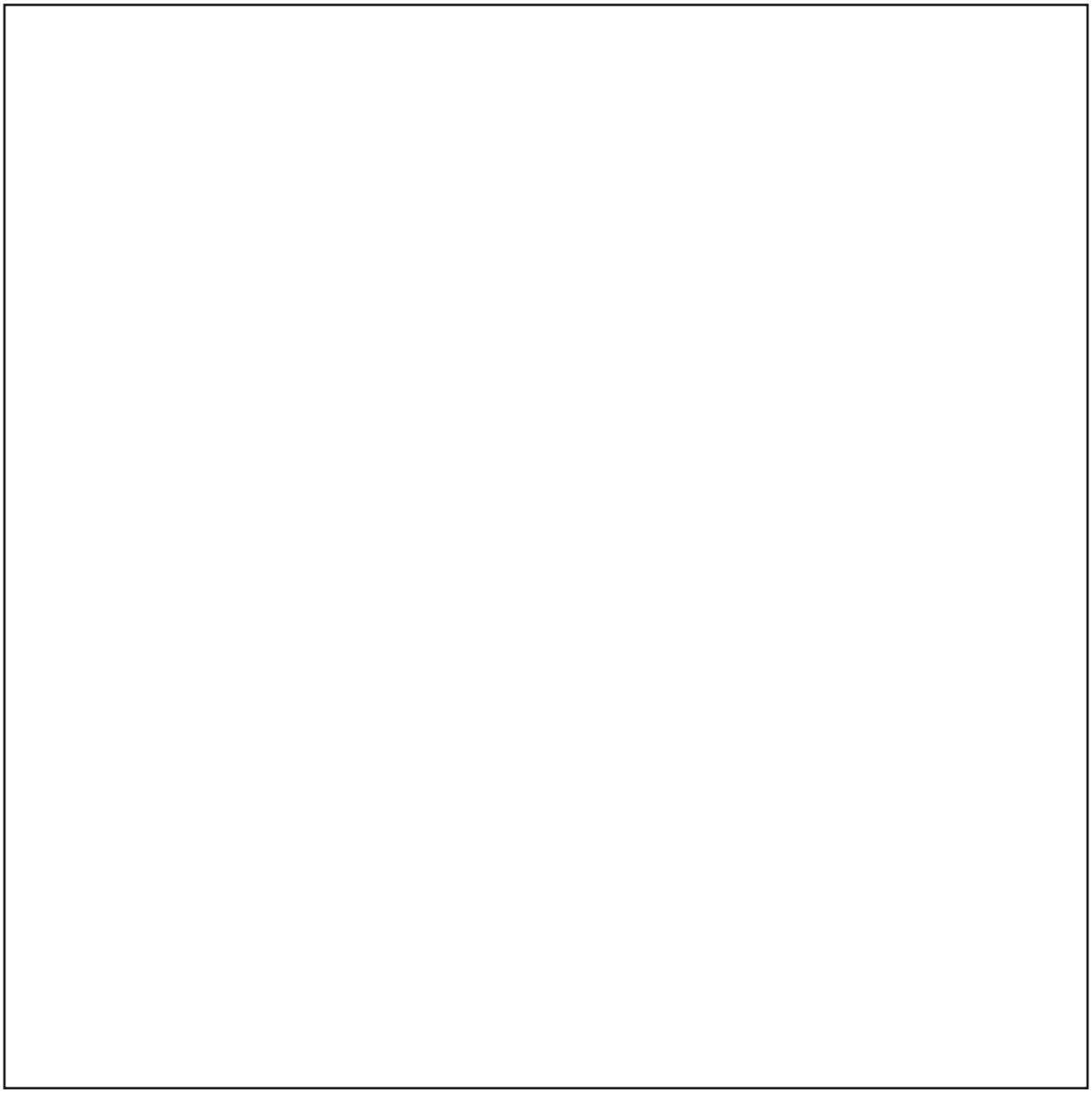


TOUCH

Digital materiality

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TOUCH Digital materiality

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TABLE OF CONTENTS

INTRODUCTION
01-0004

TOUCH&TEXTURE
02-0009
Touch 0010
Texture 0018

CASE STUDIES
03-0024
Minecraft 0026
Apple Liquid Glass 0030
IKEA Kreativ 0035

AUGMENTED TOUCH
04-0042

CONCLUSION
05-0055

NOTES
06-0057

GLOSSARY
07-0061

BIBLIOGRAPHY
08-0062

TABLE OF FIGURES
08-0066

INTRODUCTION

Growing up in 1990s Brazil I had a very hands-on childhood. With limited access to TV or computers, we spent most of our time building toys from leftover materials and objects we found at home. We would spend hours trying to reproduce the visual and material references of the world inside tiny shoeboxes, shredding plastic cups to build fluffy bushes, debating whether cellophane paper had the right color, sound and smoothness to evoke water. That early habit of translating the world through improvised materials followed me for years all the way to my studies in Fashion Design and Sculpture, where it evolved into a deliberate practice on material training. Mornings were spent surrounded by fabrics and trims, learning the difference between a mix of 30%, 50% or 95% in polyamide composition, playing with weight, cuts and folds to represent narratives through tactile and visual cues. Afternoons were spent across material workshops, learning about the softness and hardness of each: how to follow the lines of a wood's grain, bend an iron piece, carve a stone, mold clay, cast resin. It was a process of learning to read matter through touch—a language I would later find myself trying to translate into digital works.

Toward the end of my studies, this material exploration expanded for the first time into digital territories. The opportunity came as an invitation to contribute to a collective installation on humanity's inventory, in which a mix of physical and digital objects were displayed side-by-side through augmented reality. The 3D objects stand out immediately from the real ones, evoking the feeling of plaster sculptures. This idea that the digital could have a matter and materiality was highly appealing and has driven me since into a long series of experiments. It is important to notice that, although having worked with digital media before that, those 2D experiences so far had evoked in me an instant connection to Painture and Gravure. Their approach to space and materiality followed the same perspective and light/shadow principles taught during drawing and painting classes - the goal was to represent space into a flat surface and not to inhabit and sculpt space and matter themselves. That's why, being able to occupy this virtual space and explore its materials properties the same way I used to do in my sculpture workshops, spoke to me much more powerfully than any of my previous digital experiences.

My first reaction was to understand and approach 3D matter as clay. It seemed natural, starting with an initial sphere to pinch, grab and move its parts, adding or subtracting more material as needed. After getting a little more familiar with digital sculpting techniques, my experiments naturally inclined into the direction of using different shapes to carve empty spaces onto each other, similar to the way I would do with stone or wood. That got me interesting surfaces and evoked in my imagination materials different from the default one I've been using. It was when I realized that, differently from what happens in our physical world, where the material belongs intrinsically to the matter being worked with, in the digital space, material and matter are different subjects, configured independently. The realization drove me into studying digital materials more deeply, learning how to work with multiple materials compositions.

That accumulated physical and digital material knowledge eventually led me to work in a digital fabrication lab specialized in fashion and technology research. The lab's main focus was to explore



Figure 1. 3D print over fabric. Source: Personal work.

how digital fabrication techniques[1] could be used to imagine and construct new forms of wearables. Over the course of two years, I was able to experiment with hybrid materiality and apply it within fashion contexts, deepening my understanding of how digital and physical landscapes translated into one another. This raised a series of new questions as well: What happens to materiality once it is deconstructed into digital form and then reconstructed back into the physical world? How should we conceive and combine materials when an original digital surface is printed over a physical one—such as in 3D printing on fabric, in which the printed structure fuses directly into the fibers beneath it? I still didn't manage to answer many of these questions, but my experiences led me to question materiality in a deeper and more subjective way.

This thesis proposes diving into such subjectivities and investigating how Touch and Materiality shape embodiment across digital environments. It approaches digital interactions in a broad sense (from video games to screen system interfaces), because it addresses a subject permeable through all of them. It also proposes analyzing how Touch and Materials are implemented either uniformly through all systems, or in tailored solutions for each specific field. One of the first things to understand about touch is that it is composed of a combination of exploratory action and bodily awareness. We will delve further into that along the first chapter, but that characteristic means we use touch to come in contact with objects and that, through it, objects are also able to resist our body. This creates a link between us, the objects we touch and the empty spaces that define the environment we share.

Conducting haptic research has proven to be an exercise of mapping how different interactive elements, multisensory qualities, memories and gestures engage with one another. Because of that, this master thesis follows a methodology based on sensory ethnography (Sarah Pink, 2015), an approach that places embodied, sensory experience at the centre of how we make meaning of everyday life. To do so, I blend case studies analysis with personal design projects and interviews with experts in the field of 3D design to generate knowledge through sensory encounters specially (but not restricted to) tactile memory. My investigation aims to understand how interactive environments feel, sound, behave, and shape bodily experience.

Since this research makes use of many recurring terminologies that are applied with very specific meanings or interpretations, I would like to address some of the most important in this introduction. This will hopefully make the reading more fluid and some subjective concepts easier to understand. The first is the concept of synesthesia. It refers to the ability of translating qualities from one sense to another, making it possible to "see" with touch, "hear" from a smell or "feel" through taste. This translation of senses is closely linked to memory, as it draws connections between them based on our multi-sensory experiences. In this thesis context, the term is used to refer specifically to how our touch perception is reinforced by different senses.

Another concept I would like to approach is embodiment. This refers to the state of being a body, or the specific way a body interacts with and understands the world around it. The concept has its roots in Merleau-Ponty (1945)[2] theory that the perception of the world is grounded through our bodies and it is used throughout this work to refer to both digital and physical forms of immersion.

Because language around touch contains many subtle distinctions, it is also important to clarify the key lexicon adopted in this text. The four main used terms[3] are:

Haptics - related to or based on the sense of touch;

Tactile - referring specifically to skin-mediated touch;

Tactual - describing any generic touch experience;

Tangible - referring to features or objects made available through touch.

With these terms forming a common ground definition for the following discussions, we can now turn to the structure of the thesis and present how its three chapters are organized. We start by theorizing around the concepts of Touch and Texture. The first chapter draws from philosophical background to ground touch as a main element of perception and embodied knowledge, establishing as well texture as its main material expression. Here I also present the first differentiations between physical and digital materiality, creating a conceptual frame to analyze digital environments through multi-sensory logic, rather than only visuals.

The second chapter focuses on three case studies,

demonstrating in practical examples how the theories of touch and textures taken from the first chapter can be implemented. It illustrates different digital tactualities and materials behaviors, exploring the ways digital and hybrid experiences depend on multi-sensory and synesthesia to build embodiment. The three examples were chosen from diverse digital backgrounds possibilities: a video game, a UI system and a digital twin application, allowing to draw interesting connections and identify weak and strong points in each.

The third and last chapter sets out to think about future forms of touch between digital and physical systems. It analyses material and haptics exploration on my own design practice, as well as other academic design research examples to craft a definition for the expression Augmented Touch. It ends by building a framework on how to design touch in order to build hybrid embodiment beyond our traditional haptics. With this structure set, we can now continue to explore the foundations of touch and texture.

TOUCH&TEXTURE

The first chapter of this thesis is devoted to setting the foundations of the two main themes we'll be working with: Touch and Texture. These are both banal concepts, which is why they don't tend to be addressed frequently. Most of the time we make use of them in our daily exchanges without thinking about it. Bringing forth their definitions and examining their tensions and motivations will help us understand the scale of their relevance, as well as set the principles we should be working with. Since we are parting from the aspiration of designing to Touch and Materiality, it's vital to master their fundamentals first.

I start by addressing Touch, for which I go as far as analyzing Condillac's (1754) and Merleau-Ponty's (1945) works on sensations and perception. Both those theories were chosen first, because of the strength and depth of their review and, second, so we could have a look over the original notion of Touch and its semiotics before addressing its digital counterparts. After understanding how senses play and complement each other in our body, we will make our way to their abstraction into proper concepts and perceptions, which can be digitally translated. Alongside these theoretical references, insights on both concepts were also gathered through interviews with designers working directly with digital materials. Their perspectives helped refine how Touch and Textures are currently interpreted and applied within contemporary digital and hybrid contexts.

As for our second concept, Texture, I start by borrowing definitions from sculpture and fashion to anchor it first in the physical space, in the same way we approach Touch. Those two fields were chosen not only because they constitute my background in materiality, but also because both use texture as their primary medium (in fashion through fabric manipulation and in sculpture through shaping raw materials). After setting its basics in the physical, we will go into the definitions around Texture and Material in the digital space, specially in a 3D perspective. Since many logics and nomenclatures get updated within this passage from a physical to a digital rationality, juxtaposing both vocabularies will help precise their differences and refer more clearly to needed concepts in the next chapters.

TOUCH

The human body contains five classical senses: sight, hearing, smell, taste and touch. Those are more than just sensory organs to gather information, for both Condillac in Traité des Sensations (1754) and Merleau-Ponty in Phenomenology of Perception (1945) those senses are the way through which we exist as beings. It's thanks to them that our body is capable of interacting and gaining perspective on the world. Both philosophers also defend the unique and privileged position occupied by the sense of touch as that of the link between body and space. For Merleau-Ponty, that demonstrates the body's unique status of being simultaneously a subject (the toucher) and an object (the touched). He uses the example of two hands touching each other to explain this phenomenon of double sensations: "If I touch my right hand with my left hand while my right hand is touching an object, there is only one hand, strictly speaking, that touches. Always escaping totalization, my body is not merely a permanent object; it is 'that by which there are objects'"[4]. Even though one body, the hand that touches and the hand that is touched cannot perfectly merge, they remain distinct roles that continually exchange between activeness and passiveness. To Merleau-Ponty this way through which the body experiences touch is what he calls the "flesh"- the texture of existing as a being.

Condillac on the other side, views touch as the foundational sense for true knowledge of the external world. In his Traité des Sensations (1754), he compares how we assimilate information coming from each sense using the metaphor of a statute that, coming to life, is limited to experiencing one sense at a time. Through such a thought experiment, he demonstrates that the sense of touch is the first one to allow the judgement of external objects as independent from our own body. Via its movements and the experience of solidity from its own body, at first, then from others existing outside itself, the statue gains the tactile ideas of extension, figure, motion, distance, and situation. Those are the ones responsible to help differentiate itself and everything it comes into touch with. As we'll be seeing in more detail later in this chapter, these same tactile inputs are also useful in digital spaces. They help us establish our digital bodies and the understanding of what kind of interactions are accessible for us. In the VR game Tentacular (2022),

for example, players assume the role of a colossal ocean monster and must learn how to interact with the inhabitants of an island through two massive, floppy tentacles. Here, as in Condillac's statue, Touch is the main interaction that lets us understand space and establish the distinction between what is us (internal) and what are others (external). Only after that, the statue became conscious of a multitude of sensory stimuli coming from its other senses. Those, nonetheless, are unable to localize or make spatial judgments alone. They are dependent on Touch to teach them how to link their primal sensations (color, sound, odor, flavor) with the spatial properties derived from touch (hardness, distance, position, shape). Through this translation of qualities, sight, hearing, smell and taste become able to discern and interact with space and other bodies.

If we grab a river stone, for example. The moment it settles in our palm, we can feel how cool and smooth it is, how it does not respond to touch as another part of our own body would. We feel its satisfying heft - it's heavy for its size. The oval shape, perfectly rounded, makes us want to run it over our fingers, mapping the contours of the stone, discovering every detail of its surface, even the small sharp incision in one edge. All those are tactile qualities, things we can infer about the object in our possession without the need of any other sense. We bring it closer to our eyes now. We see a dark, charcoal grey with a glassy sheen coat that reflects the sunlight, everywhere except for one thin, darker line, where a little shadow lingers. A small dot appears and starts slowly expanding, changing its color to a lighter and duller tone of grey, making also the sunlight reflections lose strength. Colors, lights and shadows are qualities from the sight. Taken isolated though, without the first lessons taught us by touch, they seem very abstract and unclear. It would be impossible to infer that the little dark line with a shadow is an indication of the depth of the incision on one of the stone edges. We also wouldn't be able to compare the temperature and moisture in its surface to understand that the shiny coat implied the humidity still lingering from the river, disappearing as the rock dry.

Condillac argues that the eye itself is naturally incapable of seeing distance, position, shape, size, or movement. Sight acquires these abilities through the instructions given by Touch. Through that tutoring,

it becomes a spatial sense and progressively fuses visual impressions and tactile judgements. Moving our hand back and forth, for example, makes our sight able to correlate the shrinking or growing size of the stone with the actual distance it is from us. By touching its curves, we guide the eye to discern the corresponding light and shadow patterns that associate to its texture and shape. The result is that the eye becomes the most discerning and efficient sense, even to the point where sight's judgments are often mistakenly attributed to it alone instead of actually judgments of touch that have become habits of sight.

The same logic can be applied to all the other senses. Touch teaches hearing about spatial awareness by tackling the intensity of a sound in comparison to the distance it is from us. It teaches smell and taste how to flavor textures, shapes, moisture and temperatures. These haptics knowledge learned by each sense are the first step to the building of synesthesia, the ability to translate qualities from one sense to the other. In the physical world, this interweaving between senses becomes natural after we learn how to correlate each primary sensation. We depend on it to confirm our first impressions and form the bigger picture of a situation: a cool, smooth weight (touch) + shiny charcoal grey (sight) + wet earth scent (smell) + clacking sound (hearing) = riverbench stone. Synesthesia is also explored in digital interactions, as redundancy cues speed and solidifies our interpretation, helping us to make faster decisions and relate more easily to interfaces. In a video game, for example, a faint, low hum of water and the gentle vibration of the controller might indicate we are close to a river. If the controller's rumble intensifies as we move our digital hands, we might infer that we are holding something on them; a solid object, probably somewhat weighty. If that impression is followed by a dull, deep clacking, it corroborates that first impression and makes us understand there are in fact two objects clashing in our palm. The logic being used in physical and digital to bring those senses connections and help us interpret our surroundings is very similar. The main difference remains in the fact that, while in the physical the other senses are led by and come together around touch, in the digital that primacy belongs to sight.

Condillac's position on the physical authority of touch also reflects Merleau-Ponty's concept of

the habit as an embodied skill. For him, a habit is not a mechanical repetition, but a knowledge the body acquires. When the eye, for example, learns distance, this means the body has absorbed a new way of being in the world. This is the essence behind his concept of embodied cognition: the idea that our perception of ourselves depends on the body's physical interactions with the environment. Building spatial knowledge, in Merleau-Ponty's vision, is about perfecting the motor intentionality of the body, a tactile knowledge that allows us to respond instantaneously to the world without the need for rationally processing it.

Condillac and Merleau-Ponty agree that embodied cognition is a sensory and spatial experience, but both authors differ in their understanding of the position occupied by the body. The first, believes that the body collects sensory inputs from the external world, organizing them into concepts. For Meraleau-Ponty, the body is not a tool we use to collect sensorium data - we don't have a body, we are a body. The contrast between those theories is important to help us compare physical and digital logic concerning touch and textures. When considering our physical interactions, there is not much difference between a being and its body, the same way of a matter and a material - those are synonyms and indivisible from one another. When we go into digital environments though, most of our interactions are guided by ideas: simulations that access senses data in order to build our embodiment as digital beings (set of skills, memories, and sensory interpretations that allow you to engage and feel present within any digital reality).

In order to create that data collection of tactile ideas - an alphabet of Touch we could call - we need to isolate and abstract the properties of bodies. Even more important, we need to understand the variations between opposing tactile qualities such as solidity and fluidity, hardness and softness. Those contrasts are the tactile vocabulary that allow us to map and build our understanding of digital bodies and materials. According to Condillac, sensations come from the interaction with other sensible bodies and exist out of ourselves. In that case we can detach their qualities from their original beings and configurations and form abstractions, applying those to different bodies. These isolated concepts are called in Traité des Sensations (1754) as ideas, the memory of a sensation, touched with our body

and made into an autonomous feature. Merleau-Ponty also develops the notion of tactile memory although, according to him, the body retains not abstracted ideas but motor habits — physical memories of movements that shape how new perceptions are understood. Through the perspective of his late work (The Visible and the Invisible, 1968), he treats tactile and the digital as different scales of embodiment. The digital body, he defends, might be read as a new modality of flesh — an expanded field of our physical relations - where one can touch and be touched through images, surfaces, materials and data. A similar interpretation was expressed by Akihito Hira (innovation & digital product leader and 3D fashion specialist in Lojas Renner) during our interview for this research:

> I think that touch (in fashion design) is the first point of connection between the body and the material world. It conveys aspects like temperature, resistance, elasticity, and also this sense of memory that fabric carries. But I also think it communicates emotion and a sense of belonging. Touch, I believe, is what turns clothing into an experience. And in digital design, this dimension kind of disappears—it shifts. I think it becomes a visual and somewhat kinesthetic translation of touch, where the gaze also learns how to feel.

Mark B.N. Hansen also addresses this "becoming virtual" of the body in his work, Bodies in Code (2006) challenging both Condillac's detached body and Merleau-Ponty's "Flesh". While still following this later premise that body and being are the same thing, he shifts the focus to how this "lived body" is fundamentally changed by the digital environment, becoming something else. Hansen is concerned with how the body experiences and adapts to the digital, affecting the way we understand embodiment. He suggests, for example, that during this transition, we end up developing new sets of digital vocabulary, containing both new motor habits and new conceptual ideas. In his opinion, this new language would be able to translate physical properties to new virtual ways of touch, influenced by code and data. That's very similar to what we observed from our analysis of the physical and digital river stone before. In our physical version, we had the motor habits of grab, run over fingers, cradle in our palm, move closer and further from the body and the ideas of coolness, smoothness, weight and shape. In our digital version, the stone tangibility

was accomplished by the motor habits of pressing (screen/button), vibrating and tilting (joystick/screen/controller), and the ideas of solidity, weight, temperature and moisture.

The primordial nature of tactility in our physical reality makes touch a necessary bridge to link the physical body to the digital interface. This connection prevents the digital from becoming purely abstract or immaterial. Hansen's theory, just like Condillac's and Merleau-Ponty's, defends touch importance in understanding the digital environment. He argues that there is no pure cyberspace divorced from our physical existence: every digital interaction is already a mixed one, anchored in the physical tactility. This hybridility is what makes the digital not a space where the body's rules are abandoned, but one where we can adapt and acquire new embodiments, merging new tactile qualities and motor intentionalities. Through such combinations, we are able to build new sets of haptic skills and reflexes like swiping, tapping, scrolling, pushing,

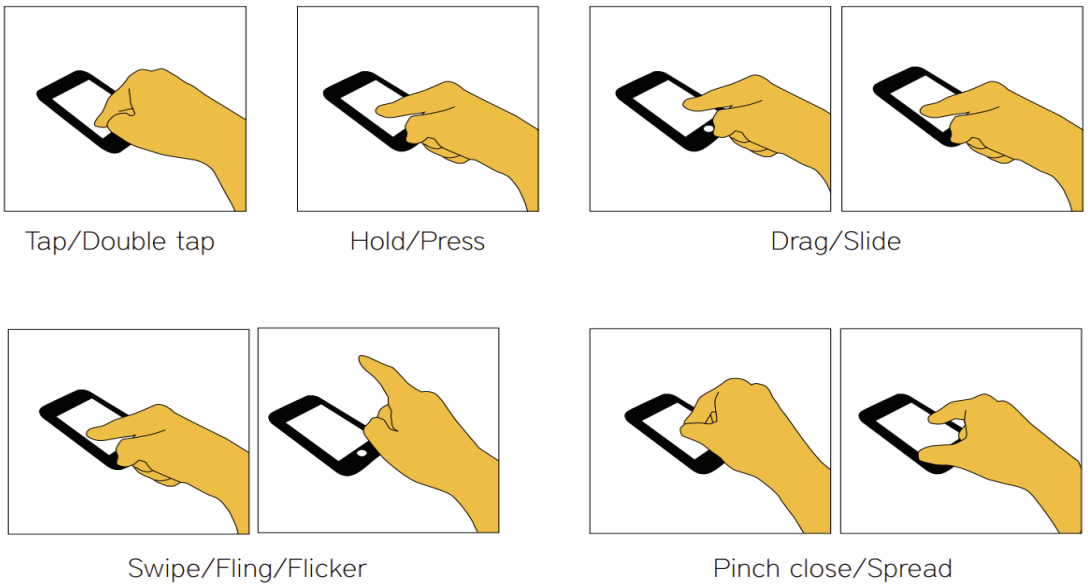


Figure 2. Motor habits linked to touchscreen gestures. Source: (NOVA et al., 2012).

leaning and combo patterns.

A further analysis from this perception of media through the point of view of our senses can be found in Laura U. Marks work Touch: Sensuous Theory and Multisensory Media (2002). In it, the author defends that human communication begins in materiality. Through the symbolic appropriation of firstness (immediate signs, perceived by the body) and thirdness (abstraction of these signs - their representation, perceived by memory and culture), Marks supports the idea of a multisensory media experience. She starts by addressing the concept of synesthesia, describing such as a translation of embodiment that refreshes our perception of an event (be it physical or digital). For her, being able to transmit an experience means to trace its symbolic meaning as well as its materiality history through different sensorium perceptions.

Marks also approaches the term haptics, central to her work. This is described in her book as a perception that combines tactility (skin mediated touch), movement, and proprioceptive functions (unconscious perception of our own body's position, movement, and effort in space). The goal is to allow us to experience touch both on the surface and on the inside of our bodies. For her, haptics in media can be experienced in different formats, not all dependent on physical touch itself. The constant flow between touch and vision (haptic and optical) establishes what she calls our two main ways of seeing: Optical Visuality and Haptic Visuality. The author explains the first as the purely visual and practical way of seeing: the vision that maintains a certain distance from its subject and understands it in a detached, symbolic way. Haptic visuality, on the other hand, refers to an embodied and material vision that uses the eyes "like organs of touch" and engages the whole body in the process of seeing. This can be found in the work « Otiosi » from Z1 studio, described by Camile De Dieu during our interview: « We did a digitalization of a forest as part of a music project, and it was about creating an archive of a place that might one day disappear. In the way things move, one could imagine feeling the wind, for example. In that case, it was purely visual, but we still make those connections in memory ». In this example, the image of the trees moving was able to transmit the haptic visuality of the soft wind blow, but many other tactual sensations contained in the same visuals didn't get through.

For Marks, that happens because haptic and optical visualities are not exclusive, but exist as two sides of one another. She defends that learning how to touch an object or media, means to reach out for its unique network of material and symbological connections. Through these, sight is constantly influenced by the potential of touch. It's the symbiosis between sensory faculties, physical body and interfaces - the objects that extend our senses into the virtual space - that establishes for her, our true virtual embodiment.

Seen through the philosophical works analysed in this chapter, touch reveals itself as a connective sense through which we are able to understand space and develop embodied knowledge. To sustain our sense of embodiment, tactility guides other senses in physical experiences, forming coherent spatial judgments. In digital environments, on the other hand, sight leads the other sensory systems through tactile memories and motor habits. This raises a central question for digital materiality: how are tactile qualities communicated when physical contact becomes mediated by vision? The next section turns to texture, one of the primary expressions of touch, to examine how surface qualities translate sensory expectations and, by doing so, build bodies and materials.

TEXTURE

Texture is the element directly associated with Touch - a group of properties that define the way something feels. When seen from the point of view of visual arts, is the quality both inherent to a material surface and that created or intensified by the artistic work. It has the intention of evoking a specific tactile feeling and emotion associated with it. Among arts subdivisions, sculpture is the field responsible for directly manipulating and playing with materiality and its textures. A sculptor can give a block of marble the impression of a soft, pliable flesh or turn a malleable piece of clay into a hard plank of wood.

In fashion, texture is closely related to the creation of patterns. Raw materials (fibers) are made into complex surfaces by using different techniques to arrange them into structured lines, loops and diagonals. This material weaving in itself is already capable of generating a new tactility, but fashion designers also work over the finished surface with cuts, sews, folds, embroidery and other techniques in order to generate textures on top of textures. Fashion and sculpture both make use of that technique of deforming a material surface to create new touchable perceptions.

Independent of the field they're being applied to, textures can present a number of different qualities. These are usually presented in a dialectical way,

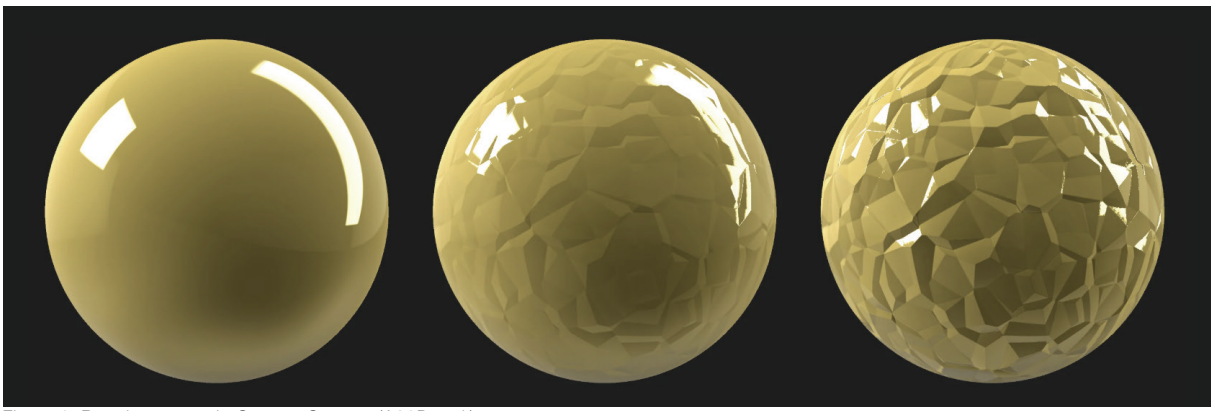


Figure 3. Rough to smooth. Source: Source: (A23D, s.d.).

through pairs that help define each other by their opposition such as: rough - smooth, soft - hard, sticky - dry, slippery - grippy, sleek - porous, rigid - flexible, warm - cool, fuzzy - pricky, moist - dry. Those adjectives are naturally referred to when describing the tactile qualities of how something feels for us. We can see many of those oppositions imported to define the configurations of digital materials as well.

When thinking in terms of materiality, digital and physical objects differentiate from each other in a primary way. If we take a 3D object as an example, by analysing it we can see it is composed of two things: a Matter - the 3D object itself - made of vertices, lines and faces[5] and a Material - the 3D object looks and behaviors - composed of images and sets of parameters. Imagine a wooden cube, for example. Its matter is the cube with 6 faces that form its shape, while its material is the combination of wooden appearance and the rigidity, density and weight of that wood. These last three characteristics are part of its behaviors: the way it will react when interacted with. The matter defines the shape of the object and the position it occupies in the space but all the information on how it will take up this place, its interactions with the environment and other objects, are defined by its material. When we digitally touch an object, it's with their matter we come in direct contact with, but it is their material we see and that responds to our interaction. When asked during our interview about the way she saw this splitting between matter and material, Holly New, VR & Digital Fashion Designer, commented:

> When you have just the 3D model, the object looks very conceptual. Then as soon as you add a texture, people start going "Oh, ok. Now I understand and see all coming together". So I think texture is really the way of communicating to people. I would say textures are realism, they bring the next level of association to a digital object.

In her work Narrative as Virtual Reality (2001), Marie-Laure Ryan addresses this significant role of communication retained by images in the digital. She explains it as another step in the subjectification (or virtualisation) of our physical concepts of reality. While digging into image representation theories, she tries to define if this movement signs we are abandoning our ties with the Real in favor of an illusion (Jean Baudrillard, 1981)[6] or if that

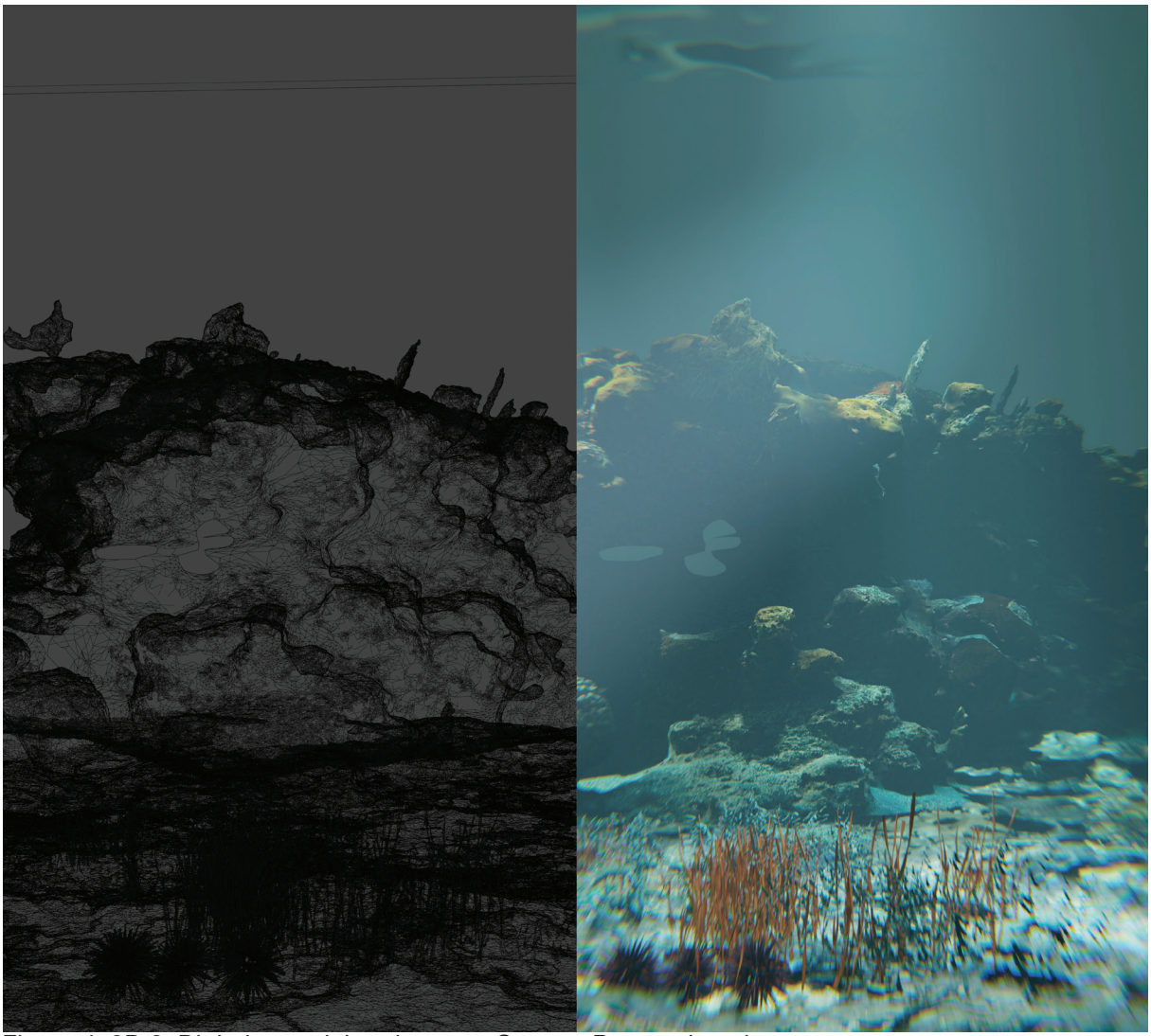


Figure 4. 3D 2. Digital material and matter. Source: Personal work.

virtualisation actually bring us closer to the "True Nature of the image" (Pierre Lévy, 1988)[7]. From the first point of view, virtual would be the transformation of all reality, from our gestures to the material substance of the world, into pure information, the Fake over the Real. From the second, virtual does not oppose the real, but nourishes it. It is another way of being that expands our creativity and makes it possible to envision new futures and meanings beneath the banal of our physical reality. Through this perspective, digital materials possess the potential to expand and enrich our understanding of reality.

In a very crude definition, a material is the substance from which things can be made. For physical reality, it is the matter that gives form, appearance, texture, and behavior to an object (meaning that all of those are one single thing). In the digital, however, a material is the appearance and behavior of an object (it does not contain matter and form, but complements it). Because of that separation, digital textures have a slightly different definition than their physical counterparts as well. To begin with, in digital a texture refers only to the visuals of an object. We can have, as in our last example, a wooden box that melts like water or inflates and flies like a balloon. The way a material looks (its texture) and the way it behaves (dynamic physical qualities) are different things entirely[8]. This also changes the way we feel digital materials. How would you define the feel of a wooden melting cube? Is it rigid or flexible? soft or hard? moist or dry? The tactile qualities are still present in their pairing oppositions but they are not fixed to a specific surface - a same texture can be assigned to different and even opposing qualities. That's what Ryan describes In her work as arbitrary physics[9]. She explains that, in digital reality (specially VR in her analysis), we are free to recombine senses in any way we want. We could build new synaesthetic relations through which "sounds have colors, colors have smells, smells have taste, and words acquire a body of their own"[10].

Digital materials have another important characteristic: each quality of their visual properties can be addressed separately. When we refer to a physical object, its appearance can change depending on the kind of finishes given to its material: for example, if it was painted, sanded or if it received varnish. With digital objects, these finishings are layers constituting the visual qualities of the material.

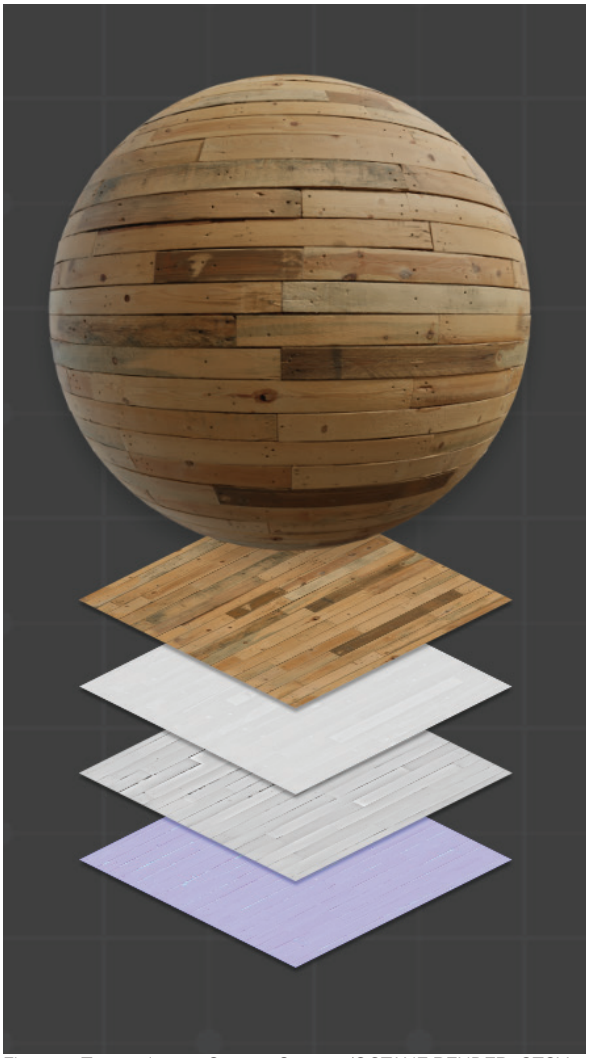


Figure 5. Texture layers. Source: Source: (OCTANE RENDER, OTOY, Inc., s.d.).

Going back to our wooden cube, this means that the wooden color, roughness, shining and relief size between grains are each a specific quality, assigned to a different texture. That way, a digital texture is not a group of properties showing how something should feel; but each one of those properties, independently. A single digital material can be composed of multiple textures (presented as image files), each one specialized in transmitting one specific visual cue to it. For our 3D cube, this means four textures: color - gradient of lighter and darker wood grains; roughness - how light make varnished areas look slick and unsanded areas look matte; specular - the strength of reflected light in varnished areas; and normal - the illusion of fine detail like grain and scratches. All those spread around our cube as layered coats in order to transmit the impression of a wooden material. This separation and fragmentation of haptic images into packets of information, make it possible to see them as a system of signs, conveying the impression of materiality by abstract ideas.

The digital fragmentation of texture into visual concepts opens us a new relationship with materiality. Digital media introduces arbitrary physics and

multi-layered sensorium cues that expand creative possibilities and push us to rethink how we feel and understand materials. As Marie-Laure Ryan and Pierre Lévy propose, the virtual does not replace the real but transforms and nourishes it. We get revealed new forms of embodiment through the feedback loops created by physical and digital. Touch remains the link between those, embodying us in each reality and letting us to reach to them through the solidity and the tactile qualities of textures. Following this theoretical exploration, the next chapter will investigate how these ideas manifest in practice. Through the analysis of three case studies we'll see how tactile affordances[11] are being implemented digitally and how materials textures and behaviors shape our relationship with digital objects.

CASE STUDIES

In the last chapter, we approached the definitions of Touch and Texture. We saw that it's through these that embodiment and interactions with surrounding objects take place in both physical and digital experiences. We also address how materials change from unified concepts on the first one, to a network combination of visual and haptic qualities, on the second. Digital touch and materials can reconfigure subject ideas and motor habits in unexpected new configurations. To understand how they operate in practice requires looking at the environments where they are experienced. Textures in digital contexts are not just visual surfaces, they mediate interactions between users and systems. Whether we touch a screen, navigate a virtual world, or manipulate a 3D of a real object, textures become interfaces that guide our actions and communicate possibilities or reactions to our contact.

To examine how textures are implemented and construct meaning in digital environments, this chapter will focus on three case studies. These were chosen from across different domains of everyday contact with the digital: a video game, a software interface, and a design application built around digital twins. The examples were selected to cover different usage scenarios and compare how touch and materials are explored through them. The two main criteria applied to their curation were their interactivity and their usage by a large number of users. Each example demonstrates a distinct way of using materials as communicative tools, from their visual textures to their behaviors.

The first case, Minecraft (2011), shows how an extremely simple visual system can generate rich tactile experiences in virtual environments. The second, Apple's Liquid Glass (2025), explores an interface (UI) digital material designed to behave like a new substance shaped by touch. The third, IKEA Kreativ (2022), reveals the tensions and possibilities in connecting digital and physical materiality. Together, these case studies illustrate the spectrum of digital texturing practices and how they shape users' capacity to feel the virtual space.



Figure 6. Minecraft world. Source: (MOJANG AB, s.d).

MINECRAFT

Minecraft is one of the most popular video games in the world with over 300 million copies sold worldwide and over 204 million monthly active players[12]. The game is a sandbox where players can build, explore, and survive in a world made entirely of blocks. The main idea is letting players interact with the different cubes they can find, each representing a different material (dirt, stone, water, etc). Players can either destroy or use the cubes to craft new materials, tools, structures or even entire landscapes. It offers two base game modes at launch: Survival mode, with a focus on gathering resources, managing health and hunger, and Creative mode, which intends to allow building freely with unlimited resources and no constraints like danger or hunger. Other variants of those can be found as well but in general the game modes change the goal of interacting with the blocks, not the core idea and characteristics of the gameplay.

The reason this game was chosen as a case study for this research is its peculiar approach to texturing. Many award-games (such as The Last of Us part II, 2020; Senua's Saga: Hellblade II, 2024; Red Dead Redemption 2, 2018; Cyberpunk 2077, 2020; Horizon Forbidden West, 2022) prize themselves for having realistic, cinematographic graphics, using v-ray and a dozen different techniques to deliver the closest to reality experiences. Minecraft, on the other hand, with its pixelated look both in its 3D models and textures, may seem an odd choice for a project highlighting textures. Nevertheless, it has one of the strongest examples of texturing usage in video games, because its materials choose to communicate with players through visually minimal cues. Those focus on the general impression and very specific details that compose a surface differentiation. This technique is observed as well by Laura Marks (2002) while analysing the similar case of the heavily pixelated images recorded by Fisher-Price PixelVision (1987): "Pixelvision, the format of the discontinued Fisher-Price toy camera that used audiocassette tapes, is an ideal haptic medium. Pixelvision, which cannot focus on objects in depth, gives a curious attention to objects and surfaces in extreme close-up. Thus some of the best Pixelvision works focus on scenes of detail."[13]. Minecraft uses a similar logic, exploring a lot the materiality of its world and expanding surface explorations to a maximum by playing with numerous recombinations of a few pixels. This makes even

the slightest color change assume a big relevance. Also, to add it up, each material in the game is a multi-behavior-driven surface, containing various reactions to the player's touch.

In the case of videogames, the word Texture refers to the visuals applied to all game objects, characters, effects, inventory and even the environment. It allows players to recognize what it is they are interacting with. In Minecraft, materials visuals are formed mainly by one single texture and not multiple overlaid images as explained in the last chapter. This is because their chosen way of passing haptic information is through minimum visual cues, expressing just enough through abstract patterns to evoke tactile ideas. These are then complemented by other behavior signals. Even with their simplistic visuals, removing the textures from Minecraft would turn the game not only into a sad and boring grayish-looking prototype, but actually render the whole experience pretty much unplayable. Texturing is an extremely important affordance in Minecraft as most blocks in game have nothing else to differentiate themselves. As so, it influences almost all of a player's choices as which blocks to destroy and which to ignore, since breaking stones (gray pattern), for example, is kept to a minimum while looking for diamonds (gray pattern with blue dots). It also helps navigation, telling players where to walk, as lava blocks might burn you (evidentiated by heart icons decreasing, flickering flame on screen edges and character and screen flashing red). Split-second important game decisions are also mediated by texture, as whether the shape walking in your direction is a peaceful villager or a zombie. All of these can be immediately recognizable by their different color pattern, but detaching these would leave players struggling to make any interaction.

The game counts an overwhelming amount of approximately 2,300 distinct textures (version Java 1.21, 2025)[14] - a number that keeps increasing as players are able to develop and share new packs of customized materials. Considering that the standardized format for these square images in the game is only 16×16 pixels, the extent of detail and variation achieved with such limited options makes the number of textures even more impressive. That wide range though, counts only the superficial changes applied to the game visuals, not its diversity in materials.

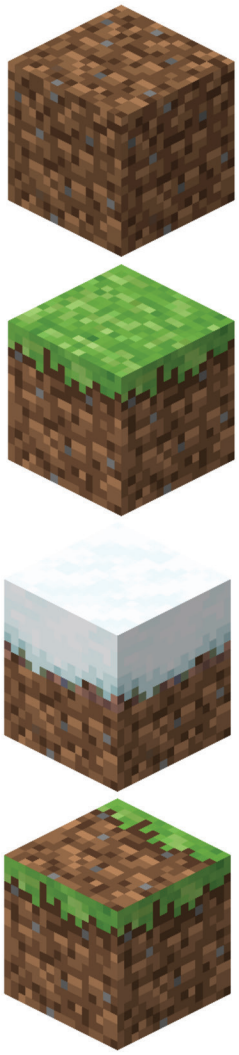


Figure 7. Minecraft grass blocks. Source: (MINECRAFT WIKI, s.d.).

As seen in the previous chapter, the sense of touch does not rely solely on a single property to identify a surface. The senses are often interconnected and supply cues to one another to enrich interactions. In the case of Minecraft (and video games, in general) these can be found in the form of audio, visual and behavior feedback. A block being walked onto will produce a specific sound, defining to which group of blocks it belongs to; one being hit by a player will sound, produce particles and, sometimes, spawn an item. Some blocks, as water, can be passed through, both producing sound and changing physics at the same time (speed of the player decreases and it gains the ability to move up and down).

Taking this holistic approach to Touch, Minecraft materials can be spread into two broad categories. One of purely decorative (look different but behave the same) and another of behavior-linked materials (associated with distinct logic, mechanics, or interactivity). In the first case, the textures (and objects they are applied to, by extension) look different but behave identically - for example, the 16 colored variations of terracotta blocks, the different 8 wood types, glazed terracotta variants, banners, carpets,

etc. This category constitutes roughly from 2,050 to 2,000 of Minecraft textures[15], meaning most materials in the game are gathered in smaller or bigger groups of similar behaviors/synesthetical reactions. Those are usually associated with one "original/default" block, as for example the oak wood (default wood) and the spruce, birch, jungle, acacia, dark oak, mangrove and cherry textures (wooden variants).

Those default textures are part of the aforesaid second case, where each texture is associated with a distinct logic, mechanics, or interactivity. Their behaviors and synesthesias can vary greatly, from the sound emitted by a plank of wood to the changing physics of a water block, exemplifying the materiality diversity explored in Minecraft. To make it a bit clearer, we will look into more detail to how we approach touch and materiality through one of the game's main materials, a grass block:

A grass block is a natural block, a cube-shaped piece of earth that forms the surface ground in most areas of the game. The top of this block is covered by a thin layer of soft grass that grows over it like a living plant material and spreads to nearby dirt blocks when exposed to sunlight, or turns into plain dirt if shaded or broken. This block's visuals (and sometimes behaviors) also change based on other environment triggers such as:

Snow - makes the grass become dormant and assume a frost-covered look;

Temperature - in warm, tropical regions it becomes lush, while in cold or dry places it becomes pale or yellowish;

Moisture - If water flows over it, the grass cannot hold and disappears, dissolving back into bare earth;

Pressure - If trampled by tools or players, the grass layer is pressed down into a beaten path, losing its soft top cover.

The grass block also produces different sounds and particles, based on the actions it suffers. When a player walks over a grass block, it can hear a soft muffled rustle; if it tries to break it, it will be presented with dirt particles (little dirt textures/images flying around the block) and a soft, dry crumbling noise. If a grass block is being placed in the world, it will produce a low, muffled thud and if turned into a path block, a quick scraping sound.

This analysis of Minecraft, demonstrates that

texturing in video games is not defined solely by visual realism, but by the successful communication of materiality through visual haptics and behavioral diversity. Materials become meaningful because they synthesize tactile impressions through interaction: the resistance of a stone when trying to break it, the fluidity of water when submerging in it, the sharp pain of heat when touching lava. The success of Minecraft's texturing system lies in its capacity to communicate through abstraction, transforming simplified patterns into perceptual tools and embodying the player in a rich synesthesia. It shows that virtual tactility is less about resemblance and more about relating to this sensorium vocabulary we all carry. This combination of multiple sensory cues, emphasizing behavior and response are also a core feature in our next example.

APPLE LIQUID GLASS

Liquid Glass (2025) is a software design language developed by Apple to uniformize the visual and tactile experience of its interface across platforms. Rather than a static visual style, it works as a dynamic, glass-like material that adapts to light, motion, and context. It refracts and reflects its surroundings in real time, flexing under touch and shifting its tint and depth to maintain clarity and focus. The idea behind it was to create an interface that could seamlessly blend with the real world while augmenting it. It works as a living and responsive material, co-inhabiting the same environment as us.

Apple's UI was chosen as an example for this chapter thanks to this particular way of re-thinking our exchanges with materials and its behaviors. Contrary to Minecraft, where textures play a role of showcasing the great variety of possible interactions open to the player, Liquid Glass does, in fact, quite the opposite. Uniformizing the whole IOS system interface to a shiny and translucent look, it seems at first glance to deprive us from the multiple textural cues that usually help guide our interactions with gadgets. It compensates by

offering instead a more subtle and tactual approach to our screens. In Apple's UX designers own words:

> "(...) rather than trying to simply recreate a material from the physical world, Liquid Glass is a new digital meta-material that dynamically bends and shapes light. And simultaneously it behaves and moves organically in a manner that feels more like a lightweight liquid, responding to both the fluidity of touch and the dynamism of modern apps."[16]

The majority of our current interactions with the digital take place through touch-screens. They are, as the name suggests, designed for touch, and their system UI's are responsible to provide us with different tactile feedback to reward us for our interactions. The use of Textures in this scenario, referring to the visual elements that allow us to communicate and control our devices and apps, represents only one dimension of those interactions. While graphics may play with depth, surface detail, and variation in color and lighting - all of which make digital elements feel more tangible and easier to identify - auditory and tactile cues follow close by in UI. They mix with visuals in order to



Figure 8. Minecraft grass blocks. Source: (MINECRAFT WIKI, s.d.).

provide clear feedback for our actions. For example, a button might have a subtle gradient or shadow to make it look raised and clickable. It might also reward the user in multiple ways, like slightly changing its color, emitting a soft, crisp click sound and a quick, subtle vibration when interacted with.

With UI (as seen with games) touch is thought in a more systemic way. It brings together different sensorium cues in order to build the simulation of tactile interactions. The logic guiding this medium however, is usually that of one element which, linked to one behavior, declenches a certain number of synesthesia reactions. Taking the previous example: a button (element) awaits the solely act of being clicked (behavior), in response to which it will activate sound (hearing reaction), vibration (touchable reaction) and color shifting (visual reaction). That said, while UIs tend to work with one behavior, it is not that rare to find materials embedded with multiple behaviors in video games. Those are used in making combinations that go beyond one-action = multiple outputs logic. If we look Minecraft water block as an example, we can see a multi-systemic material with 5 different behaviors:

Flow behavior - spreads water following gravity and other near block shapes;

State behavior - changes material properties, freezing it to ice in low temperatures or turning it to obsidian or cobblestone when meeting lava;

Environmental behavior - extinguishes fires, pushes entities, and carries items.

Ecosystem behavior - hosts fish, kelp and seagrass;

Player behavior - modify player breathing, buoyancy, and visibility.

That sort of complex, multi-behavior materials are not rare nowadays in games, but having them in interfaces mediating daily needs it's quite exceptional. Apple's Liquid Glass is built out of one single texture but it has multiple behaviours as the example given before, reactions such as flexing, changing color and dissolving backgrounds. Also, the way to unlock those behaviours is by touching the material with different gestures or patterns, exploring and building new digital motor habits. It reverses the idea of textures being used for differentiation, providing environmental cues or evidentiating disponible actions. Those abilities are instead transferred to the user. In Liquid Glass it is our touch and intention that shapes the object,

gives it identity and creates new interactions. A longer touch, for example, may detach a switch button from its layer and bring it forward to signal it is possible to change its state, feeling as if it was literally being picked up. One of the system beta testers testified: “I’ve been testing this on my iPhone (...), and I keep catching myself just... touching things to watch them respond.”[17]

Liquid Glass counts a number of 8 different behaviors associated with its single texture (Morphing, Lensing, State, Modes shifting, Scale, Tint, Blur and Highlight). Each one of those controls how the texture responds to user touch, motion, and environmental shifts. They simulate reactions that seem to position the material between a digital and a physical experience, inviting the user to embody a dubious hybrid space. In order to make each behavior clearer, we’ll take a deeper look into their characteristics:

Morphing: The material responds to user interaction by deforming its surface. It reacts like a fluid, compressing or stretching when touched, in such a way as to create smooth transitions between shapes (as when icons expand or shrink for opening or closing menus). Elements like buttons can also lift temporarily when pressed and create micro-ripples or elastic rebounds when let go, giving a strong haptic impression of spatial presence. The material produces light haptic taps or soft elastic vibrations to reinforce the impression of physical tension and release, both accompanied by soft sounds as well.

Lensing: Liquid Glass main visual characteristics it’s it refraction. Elements made with this material behave like lenses: warping, blurring and magnifying the content beneath them. Their surface also glints and their contours shimmer when the material moves, as if it could reflect light from the physical environment itself. Those are usually followed by very soft and thin chime-like sounds.

State: The material transitions between states as if it was a material with shifting phases, similar to how liquid forms or evaporates. Menus adapt smoothly to changes in their background, for example, condensating or evaporating when called or dismissed. Those movements are frequently paired soft pulses or subtle vibrations, imitating the behavior of a matter shifting density.

Modes shifting: Alterations in the background

influences the content in the upper layers, adapting it between light or dark modes.

Scale: Depending on content and gestures (like touching, pressing or scrolling), liquid elements adapt their size accordingly. When scrolling, the lower bar shrinks to give more space to content. If taped, it grows bigger and casts deeper shadows. Those resizings are also reinforced with micro-haptics and brief vibrations.

Tint: Similar to the modes shifting behavior, but adjusts elements independently (doesn't change the whole system), depending on the background colors and contrast.

Blur: Based on the motor-habit of scrolling, this behavior blurs content when this reaches higher hierarchy elements. In Liquid Glass, the content layer sits between the background and the higher hierarchy menu layers, which makes it dissolve behind these last ones when scrolled underneath their glass surface.

Highlight: When interacted with, the material illuminates from beneath the fingertip touching it to spread onto other liquidGlass elements nearby. Haptic micro-pulses often accompany these light effects as well.

Instead of using texture cues to inform about



Figure 9. Liquid glass icon. Source: (KUSWAHYUNI, 2025).

possible interactions with the system, with Liquid Glass the material's properties are shaped by the user's gesture and intention. The multi-behavioral materiality adds more visual tactility to digital interactions, playing a lot with motor habits to explore a new vocabulary that allows touch to transform the object it is touching. It seems to adopt Marie-Laure Ryan (2001) words when she says that "…the key to immersive interactivity resides in the participation of the body. (…) I do not wish to suggest that interaction should be reduced to physical gestures, but rather that language itself should become a gesture, a corporeal mode of being- in-the-world"[18].

In sum, Liquid Glass represents a significant change in interface design, where a single, uniform material can be carried with multiple behaviors that respond to touch and both, physical and digital environments cues. By shifting the focus from static textures to dynamic interactions, Apple exemplifies a new approach to interaction design; one where hybrid embodiment is built by tactile engagement. Through Liquid Glass, we can see these everyday digital exchanges with materials becoming more physical. Next, we'll take a look at how physical products address their materiality digitally.

IKEA KREATIV

IKEA Kreativ (2022) is a tool launched by IKEA to let their clients digitally plan, visualize and shop for furniture. The experience starts with the creation of a 3D digital space, where users can choose from an existing showroom or build their own room through a mix of picture and space scan. The visuals and depth information collected are then analyzed by computer vision algorithms in order to detect walls, floor planes, furniture shapes, lighting and perspective; building something similar to a digital copy of it. After these base steps, the new digital room becomes directly edible. Users can select the objects they would like to remove from the space and "clean it" through AI Inpainting, a technique in which the pixels can be recognized and covered by a masking layer of generative painting, reconstructing hidden parts of the background. Once the digital model of the room is ready, the last step is to place IKEAs products in it to see how they fit and look together. The furniture models use the previous computer calculations to sit realistically on floors and stop at walls, correct the brightness and shadows cast over them and scale the digital objects to correct proportions. Users can still

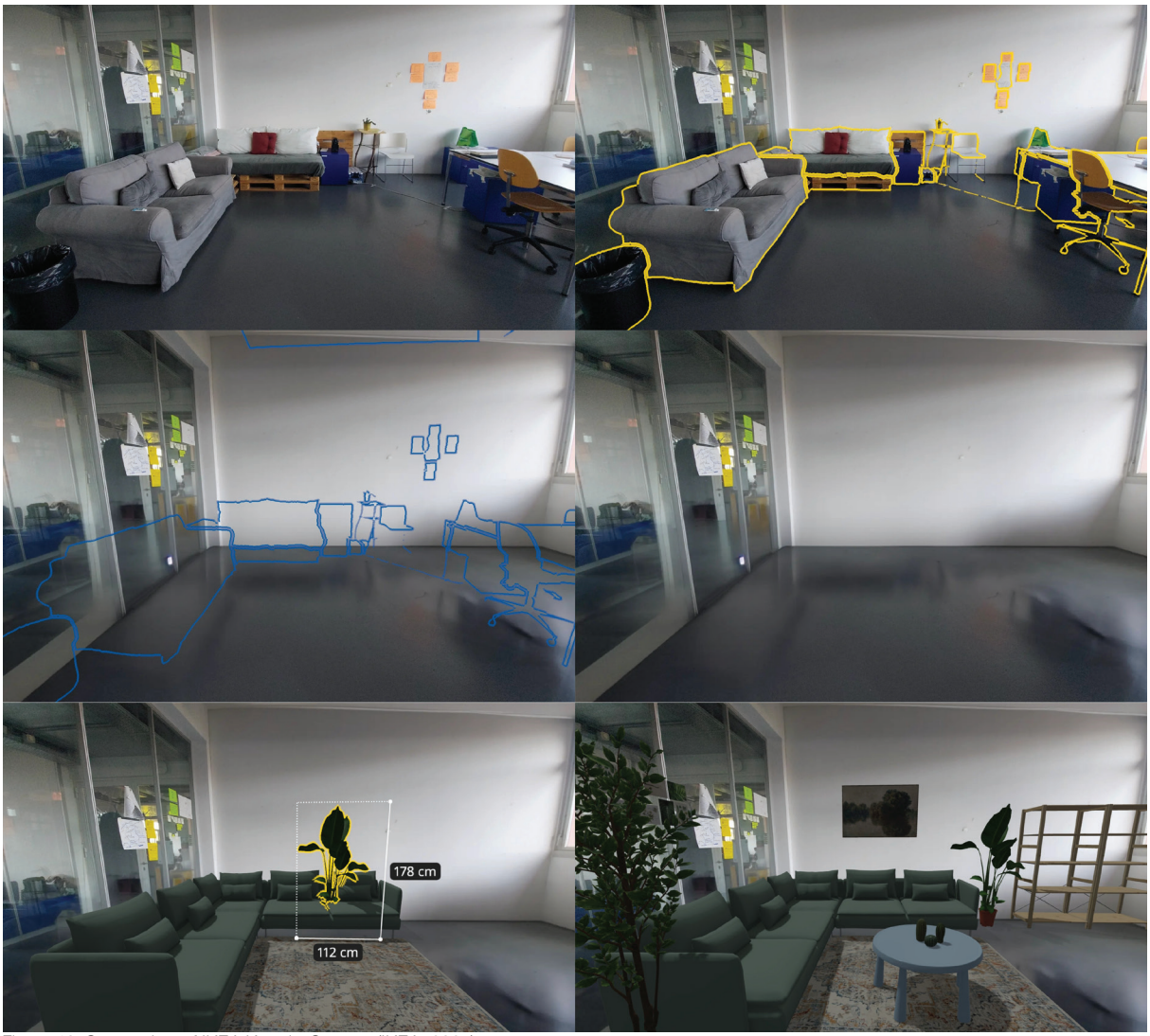


Figure 10. Screenshot of IKEA Kreativ. Source: (IKEA, 2025)

rotate, move or replace objects to see other options or different colors and patterns from its variations.

This application was chosen as the last case study of this research in order to bring a digital twin perception on the usage of digital textures and touch. It will also serve to bridge our way into the last chapter, which approaches touch and materiality in a more speculative way to rethink its possibilities. Digital twins can be defined as digital copies of a real-world object, entity or system, containing both its visuals and its properties while synchronizing data between them. It is mainly used for digital simulations of a real world application, such as: cities and buildings twins (simulate energy, transport, lightning), cars and aircraft twins (speed, fuel consumption, engine health) and design twins (materials behaviors, usage, patterns), among others. This field plays an important position in the realm of digital textures as it works closely with digital and physical materials, constantly having to translate them across these environments. As far as it goes though, most of these processes remain restricted to the design and production backstages, still rarely making their way to the general public (which takes place mostly in the form of digital try-ons). As the interest of this thesis remains in studying interactions built for broad audiences, IKEAs Kreativ approachable digital twins and physical products make it possible to analyze how those two can be perceived and compared by users.

The term texture when referring to digital twins and digital prototyping (the digital simulations of models before their physical production) have some particular properties. It can refer, as in video games, to the images applied onto each 3D product surface, giving its color and material appearance. It can also mean, in the cases where virtual try-ons are used, the visuals and lighting of real-world objects, spaces or people the digital twins interact with through cameras and pictures. At last, taking into consideration they also deal with the physical reality, it links back as well to the physical and tactile definition of texture as “the quality of something that can be decided by touch, for example whether it is rough or smooth, or soft or hard“[19]. In the Kreativ case, Texture corresponds to the images applied to the 3D furniture surface and that defines its materials appearance. It may also refer to the scanning picture users are asked to take, in order to create the digital version of their room, as well as the masking

layers used to simulate the background visuals. At last, it relates to the actual physical materials employed to produce the physical twin of the simulated products.

In that way, IKEA Kreativ states yet another way of dealing with textures and touch. The Minecraft example contained thousands of different textures, and hundreds of behaviours, and Liquid Glass worked with one texture that behaved in 8 distinct ways. Kreativ, on the other hand, includes a huge library of textures (digitalizing most physical materials contained in the IKEA catalogue), as well as bringing clients' own room textures into their activation. But those materials do not contain any behaviors intertwined with them, apart from the brightness and shadows casted over the digital objects inserted in the scanned image space. When we consider that this experience is supposed to come in hand with the user's physical interaction with the same products, something seems to be missing.

A possible reason for such a choice could be the fear that this dual reality interaction would prove itself deceiving for the client. Since engaging with a digital twin suggests at some point a touch stage with the actual physical product, building a complex digital material with multiple tactual behaviors could prove itself hard and risky. Users may have different sensorium interpretations of those and consider the comparison between both twins to be deceptive. Although a valid position, it is important to remember that many clients already return products for the same reason, because visuals-only can be very misleading if you plan on later engaging with a product in full senses.

Another probable cause could be that, as Frederic Frache (digital chain leader at Decathlon Lille) mentioned during our interview about his experience with digital twins in Decathlon, the link between digital and physical twins is sometime very hard and subtle to be kept:

> When you're working with physical and digital products or components, one is always lagging behind the other. It's a tremendous effort to keep the digital twin up to date to the physicality of what you're doing on your prototype, for example, or later when you arrive at the development phase of your product. At some point, both of them are going to separate a little bit. And in Decathlon, that's going to be at the moment

you are going through production, because the supplier is going to make small arrangements to make it easier to produce, or because the machine is not able to do that, or for efficiency reasons during the production.

Whatever the reason, especially in IKEAs case, reducing their product's tactile experience to only its visual properties seems problematic. During the initial part of the research for this thesis, I conducted two different field observations on touch, to better understand people's approach to it. That experience allowed me to collect some data that might be interesting in this case. The first exercise, focused on analyzing people's behaviors to tactile exchanges with objects in everyday activities. The second, tried to document every touched surface in a day. In the first case, an interesting observed phenomenon was that people in general avoid touching things, if possible. Apart from some designed-to-touch surfaces (such as handrails, grab poles and door knobs) most people seem to pass through places with little or no tactual contact help, relying on sight, hearing or smell to guide them. The situation would drastically change as soon as they engaged in a shopping activity though. The act of choosing an item, no matter its category (clothing, food, furniture), to belong to you, seems to instantly requisite touch to make sure the candidate is a fitting choice. Before selecting any item, people would run their palms or fingers through them, pinch them or lightly squeeze them in their hands and evaluate their weight, temperature and general comfort to, only then, make the decision if they wanted to keep it or not.

During the second field observation, I could easily document every touched surface I came in direct contact with for half a day, while visiting multiple sites. The moment I returned home, however, the number escalated so fast that I had to give up documentation after reaching 200 surfaces. Both these exercises show an interesting and unconscious approach to touch: we tend to use it in things we consider to belong to us, or to appropriate new belonging parts. That is why I consider ignoring IKEAs products' tactual properties in Kreativ as extremely limiting to clients' engagement.

It is interesting to notice that, in comparison to the other examples, what is lacking in Kreativ touch experience is not the graphic part of its 3D objects. Even when compared to Minecraft, their textures

are visually much more “realistic“ and are evoquing physical materials people have a deep tactual knowledge of as they populate their home, such as cotton sheets, fur carpets, wooden tables or stainless steel handles. The things missing in IKEA Kreativ are precisely the holistic properties of touch: the way a material is supposed to sound, move, change and react to our touch, combining to help us really feel it.

In our interview, Frederic Frache (digital chain leader in Decathlon Lille) corroborated this idea, saying that sometimes 3D created materials give the impression of having “no soul”. He explains that, while working with digital twins, there is a lot of information that is lost during materials translation, such as small physical imperfections and impressions that our brain just registers and blurs off. According to him, “Most of the time our users are complaining that our materials are quite bland visually because we are quite realistic. But they feel that they don’t look real because it’s too subtle in a digital space to be realistic.”

The analysis of IKEA Kreativ brings an interesting discussion to the role of digital materials. Digital Twins aims to simulate objects’ reactions to its users or to the physical environment they are supposed to inhabit. This creates a direct comparison between both experiences, making digital limitations highly obvious when this proposes a faithful reproduction of physical encounters. As seen in Minecraft and Liquid Glass, reproducing the real infers more than just reproducing its visuals. It should bring other behavior and synaesthetical cues such as changing the way a player is able to move when inside a water block or how an interface menu condensates when touched and evaporates when scrolled. It should also include the sound of water flowing or splashing and its constant vibration or the soft pulses indicating a shift in density in a UI element. Through those, we are able to experience physical redundancy and imperfection in enough variety to allow us to refer to the subjective and sensible idea we stored of a material in our memory. But the question that remains after those analyses is if we should build digital materials aiming to reproduce the physical. If the digital is a sandbox where the rules from our physical reality can be bent and recombined in any way we want, wouldn’t it be more useful for us to use it as a tool for imagining what could exist, instead of what is already here? This proposes the digital can have the capacity to overlay digital information

onto physical practices, an investigation that will be better explored in the next chapter.

AUGMENTED TOUCH

In the last chapter we analysed three case studies to identify how touch and digital materials are seen and worked with today, through practical examples. We saw how Condillac's concept of ideas and Merleau-Ponty's concept of motor-habits are still actively being applied to evoke embodiment into digital and hybrid experiences and could observe how these tactile experiences add to our perception. We also started questioning what should be the position occupied by digital touch and materiality in our exchanges, specially in hybrid environments.

This chapter is dedicated to building this speculative notion of what an augmented touch could be. It starts by presenting some of my own work with digital and physical materiality, showing how this expanded notion of hybrid touch evolved through my career and combining those with excerpts of interviews with professionals working with digital translations. The last part introduces other academic design projects dealing with digital touch and brings together the theories and examples from the last chapters to define the better practices for pursuing this augmented form of materiality. We conclude by resuming all these theories and observations into a comprehensive set of guidelines for digital touchable experiences.

During this research, the one question I asked all my interviewees was what they imagined an Augmented Touch could be[20]. I initially built the expression by borrowing the same structure used by Augmented Reality, a term coined in 1992 by Thomas P. Caudell and David W. Mizell[21]. By the time, the term was used to describe a digital display system that could superimpose digital diagrams and instructions to the physical space, augmenting the view of the real world. I wanted to start from the same notion of a technology that overlays new layers of digital information with the goal of not replacing, but enhancing our experience with materiality. An augmented touch, in this sense, would belong to a broader field of enhanced reality - a hybrid way of overlaying digital qualities onto real materials, enabling new forms of embodied perception and interaction.

Throughout my early academic trajectory, touch remained a central piece of my design and artistic practice. It accompanied me from a tactile and material education shaped by sculpture and fashion design to my first experiments translating matter into digital qualities. Discovering that, in digital environments, material and matter can be configured independently, exposed new creative possibilities, as well as sensory gaps to it. When touch is no longer defined by physical resistance, nor tactile patterns, we need to start getting fluent in its network of meanings and associations. The tactual hybrid vocabulary is redesigned through combinations of new and old motor habits and more abstract memory connections, such as synesthesia. This transition from hands-on making to digital modeling and fabrication marked the beginning of my investigation on multi-materiality conversion.

It was within this context that my experiments evolved into the project UMBRA (2020). Developed in the context of isolation imposed by COVID-19, the research proposed a reflection on the boundaries between physical and digital realities. It approached virtualization of daily actions by seeking to create a “daily life texture” — a hybrid surface that captured the blurred boundaries between online and offline existence. This texture was generated by overlaying multiple daily images, combining same moment stills of smartphone screenshots and camera pictures into mappings of daily-life experiences. The resulting visual layers — merging fragments of physical surroundings with digital-only interfaces — were interpreted as material topographies and translated into three-dimensional. The textile prototypes that resulted from it, made use of a combination of digital fabrication techniques such as 3D modeling, 3D printing, laser cutting and textile manipulation.

UMBRA explored touch not through direct physical interaction, but through an evoked, memory-based tactual experience. It didn’t aim to imitate reality or produce perfect material replicas, but rather expand and reinterpret touch. To do so, it used a mix of visual haptics (creating first textures perceived only through sight) and physical touch (reconstructing this digital daily texture through digital fabrication techniques). The project proposed a form of augmented touch in which ordinary digital elements, such as buttons, menus, icons and webpages, could be made into material experiences

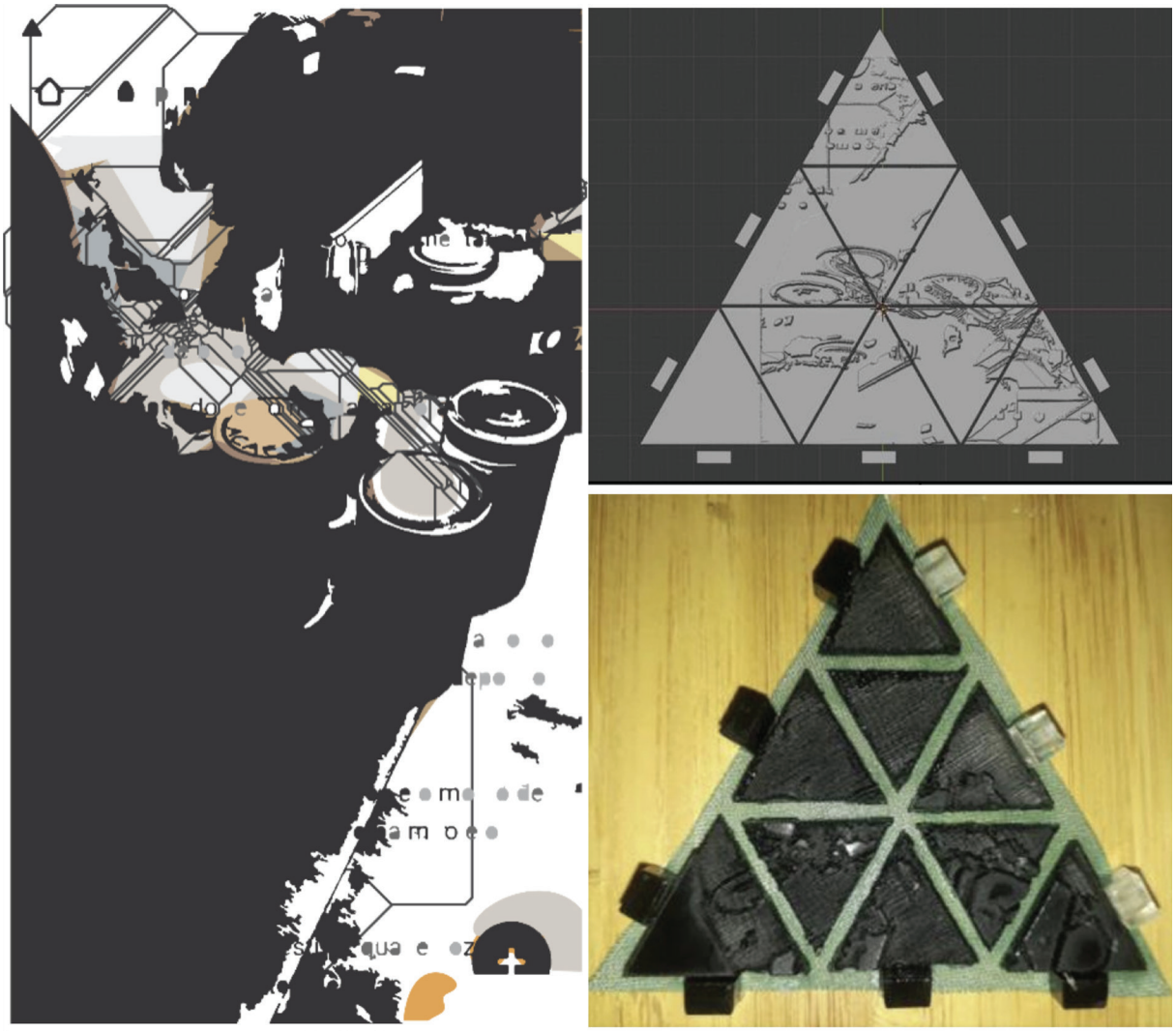


Figure 11. Umbra. Source: Personal work.

and physically handled and felt. It reinterpreted digital traces as material surfaces.

This translation between physical and digital can be approached by another perspective, through 3D experiences for marketing campaigns. In many of these cases, digital tangibility is achieved by the digitalization of physical materials. Those provided me with a great opportunity to extend hybrid materials exploration, especially through digital twins projects. One example was a fashion film produced for the 2023 Inspiramais fair in Porto Alegre, Brazil. The Inspiramais Fair is a biannual Brazilian design event that showcases innovative and sustainable materials for the fashion, footwear, furniture, and accessories industries. Many of the materials presented in 2023 were handcrafted, proving demanding to model and texture digitally. While the meticulous visual result of these digital twins was able to optically evoke the original textures in the end, the models seemed short in terms of the material richness they were supposed to carry. The digital twins looked like the originals, but didn't feel like them. They could not convey the tactile properties of tension, stiffness, thickness and fragility of the materials; nor could they transmit all the physical work artisans had invested in their creation.

The process of translating handcrafted physical materials involved long hours of visual interpretation of tension, grain, weight, ridges and deformations. It teached me new ways of observing and experimenting with digital surfaces but, most importantly, it exposed a critical limitation in digital material translation: the gap between visual fidelity and embodied tactility. The experience evidenced the "missing" holistic haptics features, the same impression we were left with in the case study of IKEA Kreative.

This feeling of frustration was stated in many of the interviews done for this project. The interviewees were mainly fashion and product designers, professionals who work mainly with digital twins, and need to constantly compare physical and digital properties of materials. For them, the freedom in creativity and the economy in ideation time are two big advantages in the use of 3D on their workflow but material simulation stands as the main reason for its adoption. Haptic feeling is crucial and, at the same time, underdeveloped to support full thrust on the conveyed feel of a product. When I asked about his impressions, Akihito Hira (head of

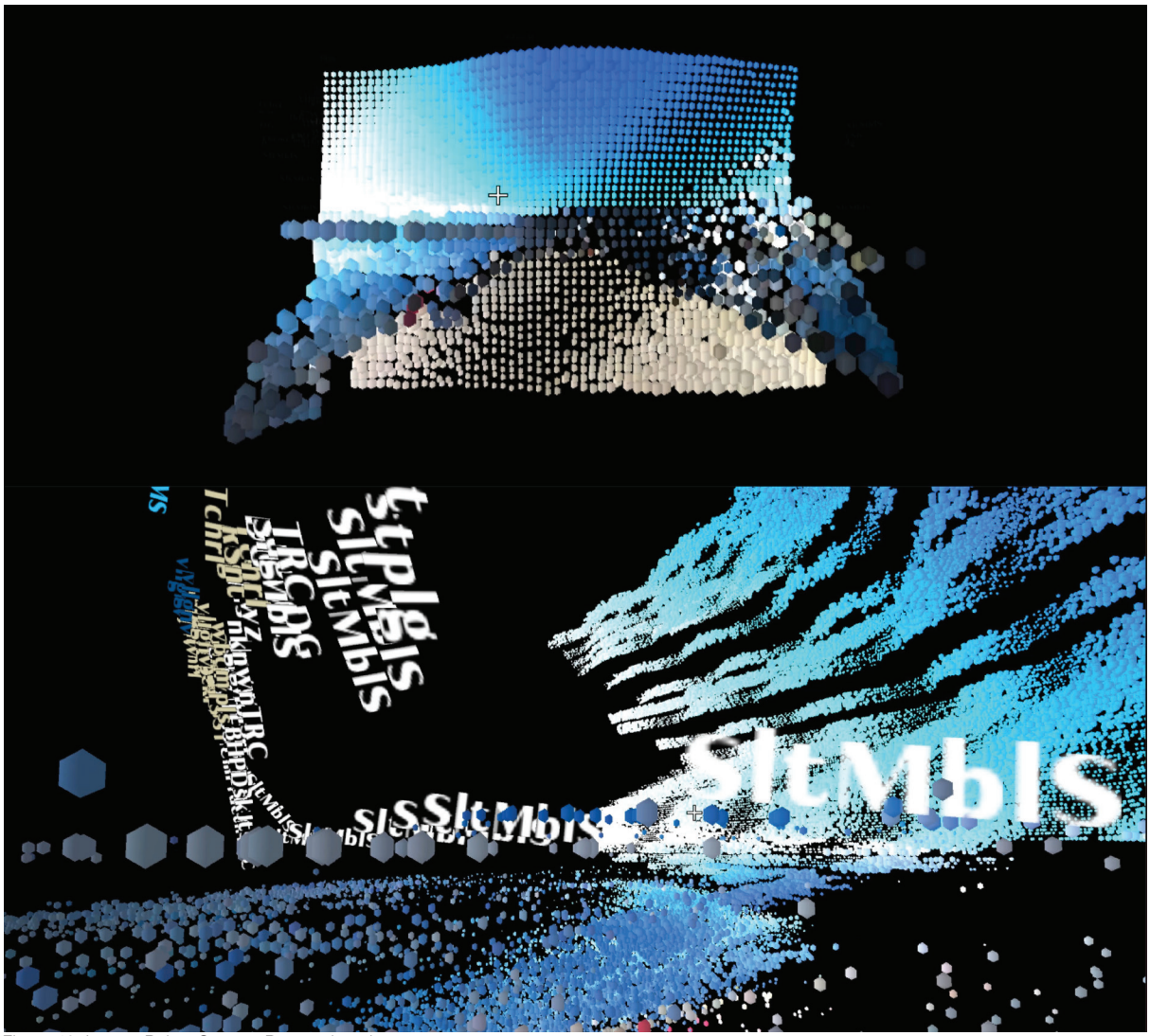


Figure 12. Leman Pulse. Source: Personal work.

Fashion Innovation in Lojas Renner) commented that for him, 3D has brought numerous benefits to the design process, specially a certain predictability regarding what will be produced. He explains his work as a dual step of crafting digital surfaces - to capture material appearances - and tailoring their mechanical behavior - to reflect those desired in the physical product. For him, someone who designs products in 3D needs to be as well an expert in textures and hands-on physical materiality. In order to be able to create a meaningful relation between physical and digital, he explains, you need to have a deep understanding of the qualities being gained and lost in both:

> During material translation, I think that, first of all, we lose the unpredictability of the material itself. The friction and, especially, the immediate response of the fabric — that sense of manipulating it, touching it with your hands — are also lost. But I think we gain a lot too; we gain a new layer of control and experimentation. We can manipulate physical properties, test behaviors, or even imagine materials that don't yet exist. I believe this notion of digital touch doesn't replace the physical one in any way. The key word here, I think, is that it expands and reinterprets all of this. Through that, we're able to explore new possibilities within the digital realm.

The gap between visuals and behavior or other sensorial cues in digital products is one of the main difficulties in building augmented materiality. In the interviews with Gabriela Scanferla (3D Creative Lead and Project Manager at Metakosmos) and Frederic Frache (Digital Chain Leader at Decathlon Lille), both endorsed that fact. They affirmed that when designing, the visual aspects of their products don't usually pose many problems, but they frequently spend lots of time making them feel right for their clients. Gabriela explains that in physical experiences we are used to empirically understanding many tactile characteristics of how products feel against our skin but these translations get extremely subjective and hard in the digital. In her work practice, she says, she usually asks her clients for visual references not only of the product they want to develop but, most importantly, of the subjective impressions they are trying to pass.

The challenge of communicating felt experience into digital representation was one of my focuses while developing the project Léman Pulse (2024). In it, texture and space were addressed with the

intention of creating a straighter connection between matter, material and embodiment. The project began as a field investigation into how people communicate through visible and invisible layers of information around lake Léman, in Geneva. Starting from interviews with shared stories and impressions taking place on its shores, the lake area was documented by photo, video, audio, electromagnetic activity (EM), Wi-Fi, and radio frequencies. This collection of human, environment and digital traces was then transformed into an interactive 3D representation of the physically visited places. Their visuals would dynamically distort based on the visitor proximity (zoom in into the space) and the intensity of invisible communication at the original location. Visitors to this soundscape could hear of the original sounds and stories, while watching the paysage texture distort around them.

Léman Pulse works primarily with spatial and imaginative touch rather than direct tactual contact. The project creates a sensorial environment through



Figure 13. L.O.U. Source: Personal work.

which touch can be explored by spatial gestures (zooming in, zooming out, rotate). The texture and materiality of the world itself is highly subjective, made of multiple new digital haptic qualities such as electromagnetic waves, wi-fi and bluetooth signals, water, wind and birds sounds, as well as stories. This overlays of invisible communication onto familiar geography, forms a materiality that cannot be grasped physically but is nonetheless part of an embodied experience.

This subjective approach to materiality, which not only engages other senses, but also builds connections with memory, seems somehow to bring us closer to the sought feeling of augmented touch. Camille De Dieu told me about a similar perception in works developed by her and Laurent Novac at Z1 Studio, in which the cues used in installation's scenography served not to reproduce every detail realistically, but to leave space for the evoked feeling to be called forward by memory and imagination. In one of these examples, a research project around melting snow, the idea was to create archives of ice that could immortalize the ephemeral existence of melting snow. Apart from the natural aspect, it should also compile the culture, knowledge and vocabulary built around this disappearing element. The main question behind the work was how to be able to talk about and transmit all the holistic concepts of snow when this would no longer exist. For Camille, translating physical or conceptual qualities into the digital requires choosing something tangible to serve as a material anchoring to the environment or experience being built. Texture, she says, " is what still brings a certain humanity to what exists in 3D, in the digital world. It reminds us of memories, makes us think of something else. And by doing so, it inevitably evokes different ways of touching as well."

This capacity for texture to evoke memory and emotional connection, grounding (or embodying) through touch, is one of the main appeals for exploring augmented touch. As commented before, the creative possibilities offered by the digital are a big attraction to exploring new contrasts and playing with our own senses. It allows us to work with synesthesia, design new gestures and propose different connections between those - new forms of feeling and communicating with other bodies and the environment. Layers of Understanding - L.O.U. (2024), is a research that brings many of

these characteristics together, even not being a digital experience. L.O.U was created from a design research I developed on skin. The idea was to develop a soft robot that could react to movement and reshape space, acting as an organic floor surface. It proposes a new speculative relationship between residents and their home as it unfolds in different shapes depending on how people approach it. It can be walked over, but only if it allows you to; it can be petted or caressed, if you stay aware of how you move around it; It can block your path and make you redo your steps if you startle it. This project took a step back from the purely digital representation to focus on the notion of touch and texture as interfaces between us and our domestic spaces. The exploration started by documenting and reimagining everyday household objects as living "skin species" — entities with their own textures, rhythms, and behaviors. Each was described through its tactile and emotional qualities — roughness, fragility, warmth, or adaptability — and its capacity to communicate with other species and environments. The project also mapped the different textures found in the body and the different gestures that evoked the notion of skin. By the end, multiple patterns were cutted, inspired by the different body textures and



Figure 14. Materiable. Source: (MIT MEDIA LAB, s.d)

exhaustively tested in order to evoke in the public the desire of touching - caressing their fingers over the robot surface. The result was a piece that played with space delimitation, movement (both of the robot and the ones interacting with it), and longing for connection.

In some ways, LOU has many qualities central to the notion of Augmented Touch we are analysing, even though it doesn't show a direct exchange between digital and physical environments. For firsts, it puts spatial exploration in the foreground, amplifying our embodied perception: everyday gestures (walking, stepping, running, caressing) acquire new meanings and consequences, revealing the body's role in shaping its surroundings. It also invites and reacts to touch, creating a form of tactile communication that resorts a lot in memory to draw associations with known motor habits to propose ways of engaging with this new entity/material. This example shows how touch can be augmented even in physical spaces by adding a new layer of interaction to affective relations, multisensory engagement, spatial exploration and everyday gestures.

From this standpoint, I believe it becomes relevant to add to our analysis a quick overview on how digital systems can achieve a sense of digital materiality through direct haptics feedback. I would like to briefly mention two research projects on that line, FeelPen (2023) and Materiable (2016). Although more technical than the other examples viewed so far, they also focus on the interactivity between digital matter, material, body and space and offer interesting views on the possibilities of rendering digital materials through touch.

In FeelPen (2023), Bence L. Kodak and Yasemin Vardar, propose a haptic stylus capable of rendering tactile properties of images. It integrates three main haptic modules to simulate the touchable properties of roughness, friction stiffness and temperature, which are called upon as the user explores flat screen interfaces. The pen can identify if a surface is supposed to be soft, stiff, elastic, smooth, slippery, bumpy, warm or cold, in a way that pressing it onto a touchscreen transmits the feeling of pressing it into rubber, plastic or wood. Among the applications of the technology, the authors specifically mention the potential of digital materials creation, empowering designers and artists to sculpt not with visual haptics, but with texture itself. Similar to the project UMBRA (2020), FeelPen

proposes a way to turn digital elements tactile. This relocation of material attributes back to haptic signals, don't infer a perfect translation between physical and digital touch. Its focus is on enlarging the number of tools available for augmenting it, giving us a new hybrid category of tangibles.

The second project, Materiable (2016), developed in MIT Media Lab, presents a TUI (touchable user interface)[22] that aims to represent dynamic material properties using a pin grid. It integrates different sensation techniques, including haptics and visual feedback, to create a model of digital matter itself. The deformable and dynamic material change its properties and respond to touch, reproducing the properties of flexibility, elasticity and viscosity. Materiable takes a step further into hybrid materiality by blending the properties of digital matter and material into something that is neither a physical material nor a digital one. It possesses a physical body, but it is still configured as a digital matter, which means it will keep digital creation potential of recombination between those two.

Those two examples speak powerfully to me, linking back to my background in materiality translation. Materiable brings the same questions I first encountered in my experiments in the digital fabrication lab, while trying to turn into physical, digitally native designs. While FeelPen seems to stand somewhere in between the digital twins material translation and my first digital sculpting tests, trying to apply traditional artistic techniques to digital matter. I believe that their shared ability to render digital properties into physical space holds essential value for the idea of augmented touch, as they do not attempt to perfectly replicate physical materials, but explore how digital properties can acquire their own sensory expression. They expand the vocabulary of what touch can be: not a substitution of physical sensation, but an augmentation that generates new motor-habits, ideas and possibilities of matter-material combinations. This perspective reinforces the idea that Augmented Touch is less about reproducing reality than about designing conditions for new forms of tactual meaning to emerge.

As we make our way to start conceiving these new haptic formats, one last differentiation remains important to be addressed: Physical touch spreads to every body we interact with, independent of their design - digital Touch, on the other hand, is always designed. This means we should also understand

and filter this tangibility aspect in augmented touch, as not everything we touch in the physical world is meant to have its solidity directly translated to digital and hybrid experiences. When we talked about this subject, Cyril Dellenbach, QA support artist at Adobe Substance 3D said that the most important thing in his point of view, is that haptic feedback should have a very clear and useful application. If it doesn't add to the user experience or if it's even slightly burdensome to access or use, people will not adapt to it. The same position was supported by Holy New, VR & Digital Fashion Designer in our interview. She defended that touch solidity can open a whole new realm of possibilities in digital experiences, as she said: "I genuinely think that, if it's not adding to the experience, then it shouldn't be in there. When I'm designing, for example, I don't necessarily want to be touching a mannequin and want to be able to move through it. And sometimes that's what you need to be able to do. You need to be able to go into a mesh. Also, when I touch a digital object, what is it that I'm feeling? Is it its weight? Its texture? Can I be touching and feeling its color? Is it blue warm or is it blue cold? At that point we begin to develop a new language and then, in order to be able to communicate through it, we need to choose the right time and place for placing touch". These perspectives show that, in addition to creating new textures and quality recombinations, ultimately, the successful application of Augmented Touch lies in its utility and deliberate application as a new layer of communication rather than mere technical application.

Throughout this last chapter, we draw from projects and professional interviews as well as previous concepts and case studies, to achieve a definition of Augmented Touch and understand its applications. We began by defining Augmented Touch as a technology-driven addition that overlays new layers of digital information to a physical practice. Its main goal is not replacing, but enhancing our experience with materiality. This expanded notion of touch has revealed throughout this chapter, five core pillars to it. The first, is its definition as a design linked to spatial exploration. This can be observed in L.O.U. (2024) for example, where the way one moves, shapes the environment and gives new meanings to haptic gestures (e.g walking, running). The second point is that it must evoke embodiment shared between digital and physical at once, as in how digital traces got transformed into physical textile surfaces

in UMBRA (2020). The third pillar is that Touch is used as its interface, similar to how FeelPen (2023) receive tactile feedback from screens. The fourth point states it should be able to dynamically render physical materials into digital information and digital properties into physical sensations. This is explored in Materiable (2016) when hand pressure renders elasticity simulation. To conclude, the fifth and last pillar is that these translations should not be only visuals, but able to work with dynamic synesthesia. This allows new memory associations and creative reconfigurations, as experienced when topography gets distorted by sound, spatial position, colors and memories in Léman pulse (2024).

These five core pillars are important from a design point of view. They provide a comprehensive framework and set of guidelines for creating meaningful Augmented Touch experiences. Ensuring that haptic technologies are not used just as gimmicks, those characteristics focus on true improvement to human-material interaction. They also allow space for new alphabets of motor habits and abstract ideas to be developed. The main conclusion taken from it is that Augmented Touch is not about replicating physical reality, but about expanding our embodiment experiences through creative recombination of materiality.

CONCLUSION

This research has explored how touch and materiality shape embodiment in digital environments. It showed that, while physical reality grounds our perception primarily in touch, digital spaces are often set to prioritize sight. When applied to digital experiences though, tactuality and multi-sensory cues sum to create meaningful interactions and leave a strong sense of presence in their traces. Through the use of sensory ethnography methodology, the work observed multiple examples of how our physical understanding of touch and texture shape the way we experience and interpret materiality.

Touch and texture foundations were established using key philosophical ideas like Condillac's and Merleau-Ponty's concepts of Touch and Embodiment. These built an understanding of sensorium knowledge for spatial recognition, showing that both physical and digital interactions depend on synesthesia to ground user's interpretation. This approach, supported by Hansen's and Marks's work on the virtualization of the body, opened new possibilities for digital embodiment. This started by identifying a core difference between digital and physical materiality: while in the first, both concepts are indivisible; in the second, Matter (geometry) and Material (appearance and behavior) are independent. These previous assessments were tested and confirmed while examining three case studies—Minecraft (video game), Liquid Glass (design language), and IKEA Kreativ (digital twin)—revealing that focusing experiences only on visual fidelity, leads materiality to a lack of multi-sensory realism. Building on this, the final section introduced Augmented Touch to speculate on new hybrid (digital-physical) interactions, arguing that design should engage with embodiment by overlaying digital layers over physical materials, based on a five-pillar framework: spatial exploration, embodiment, touchable interfaces, dynamic properties rendering, and dynamic synesthesia. To illustrate how these thesis ideas might materialize an augmented touch design through a tangible interface, I would like to propose in the following, a speculative scenario.

On a cutting mat placed over the table surface, a TUI (touchable user interface) lies, discreetly. When our fingers press down, the mat surface activates, waiting until a piece of recyclable matter is positioned on top of it. The interface instantly applies digital resistance to it, transforming the passive

material into a rigid, cool-to-the-touch ceramic. We run our fingers over it in a known gesture pattern and the TUI dynamically simulates the properties of a completely different material, with digital 'softness' and 'tension' being applied to specific areas. We keep "sculpting" the complex shape: a fluid drape scarf with a frozen rigid wave; applying the right amount of pressure to trigger the surface resistance, while gently caressing the edges to make them malleable.

Satisfied with the model, we run our fingers along its border, "scrolling" through the material library to test the feeling of a 95% cotton (elastic, smooth), a clay plate (rigid, rough) and a sandcastle made last summer (soft, rough). We touch different sections of the matter, applying our favorite textures, feeling the "digital grains" on our fingertips shift with each movement. When the tactility is done, we tap the surface a couple of times, assigning a Symphony No. 2 to the rigid curves; and change the local micro-climate of the soft edges for a slightly warmer temperature.

In a scenario like this, we can see a future in which matter and materials oscillate between physical and digital, combining those in a symbiotic way. We could imagine such encounters being guided by augmented touch and designing a foundation for a new embodied experience.

Ultimately, this thesis demonstrates that touch holds a fundamental importance to the way we inhabit and see our reality. By designing touch and materiality interactions, we are able to open new creative possibilities of rethinking and reinventing the world. While reproducing reality is technically impressive, it doesn't hold the notion of embodiment and affective connection to an interaction by itself. Touch is originally the sense that brings together and teaches others about spatial possibilities, enabling interaction. Augmented touch can though be used in the same way, to build a connection between physical and digital materiality.

NOTES

1. Definition in the Glossary section

2. (MERLEAU-PONTY, 1945)

3. (FULKERSON, 2024)

4. (MERLEAU-PONTY, 1945, p. 42)

5. 3D meshes follow the same basic geometry laws: two vertices (points) form a line, three lines form a plane (triangle)

6. (BAUDRILLARD, 1981)

7. (LÉVY, 1998)

8. There are ways to link the properties of looks and behaviors into one material file (it is possible through Substance Designer, for example), but that is still rare and both still need to be set separately.

9. (RYAN, 2001, p. 78)

10. (RYAN, 2001, p. 79)

11. (GIBSON, 1986) The term “affordances” was coined by James J. Gibson in 1977 and used to define the set of potential actions an environment, object, or interface offers to an individual. In the context of this work, it is specially used to define the designed possibilities and information given on digital interactions

12. (HOST HAVOC, 2023)

13. (MARKS, 2002, p. 11)

14. That number is a combination of an estimation made by the community and a manual textures counting in minecraft fandom catalogue (REDDIT, 2022) and (MINECRAFT WIKI, 2025)

15. There is no official published count of Minecraft textures that correspond to mechanical behavior rather than visual variation. Based on classification of block and item functionality using Minecraft Wiki registries, an approximate range of 250–300 mechanically distinct behavior categories can be identified, representing around 10–15% of all textures in the game. (MINECRAFT WIKI, [s.d.])

16. (APPLE DEVELOPER, 2025, 0:41–1:04)

17. (BHATTY, 2025)

18. (RYAN, 2001, p. 286)

19. CAMBRIDGE UNIVERSITY PRESS & ASSESSMENT, 2025. Texture [en ligne]. Disponible à : https://dictionary.cambridge.org/us/dictionary/english/texture [consulté le 19 octobre 2025].

20. Interview responses to the question: what does the expression “Augmented Touch” evokes to you?

Akihito Hira (innovation & digital product leader and 3D fashion specialist in Lojas Renner):

Ah, I think it really evokes this idea of fusion — both with tactile memory and with the digital response. Touch — or rather, touch expanded through technology — brings together matter, sound, and vibration; all of these elements merge to create new sensations that are not necessarily imitative of the real, but instead a kind of touch that is not limited to the skin. As I mentioned before, it becomes a language in itself.

I truly believe that touch will become something hybrid and, at the same time, almost ubiquitous. These tactile interfaces will be able to reproduce sensations through haptic stimuli, but we will also have many cognitive textures — surfaces that communicate directly with the brain through visual patterns and even emotional responses. I think we’ll end up wearing the digital as if it were a second skin. And I believe the sense of touch will become simultaneously physical, emotional, and informational.

There are already some experiments and attempts, especially in academic contexts, involving gloves that can be worn and make contact with the skin and fingers to simulate the sensation of fabric. (...) The medium could be through such mechanisms — gloves, full-body suits — capable of transmitting these kinds of sensations, like in science fiction films. That would be amazing, truly ingenious. But I still like the idea of imagination — of visualization. I think that brings a multisensory feeling to our minds. The visual has enormous power for us. So, for now, I don’t think I’m really missing that tactile medium just yet.

Cyril Dellenbach (QA support artist at Adobe Substance 3D):

I immediately think of VR and haptic gloves. Also, I don’t remember where I heard about it, but someone told me about a smartphone that has mini receptors with vibrations that make you feel the object or the texture you are touching. Like, if you see an object on Amazon and for some reason you want to feel it, then this smartphone can give you a haptic feedback of what it feels like. Personally, what I would love to see in the future (specially in Substance Designer) is an optimized way to work

with procedural materials. I think it's awesome to create animated materials that can change and respond to interactions.

Gabriela Scanferla (3D Creative Lead and Project Manager at Metakosmos):

What comes to me is the sensation of skin-to-skin touch — as if I could feel touch itself, but at a much higher intensity. Actually, the first thing that I thought of was an episode of The Big Bang Theory where Howard built a virtual kissing prototype. In order to be able to have a virtual relationship. He creates a pedestal with a mouth and a tongue, an interface that allows two people to kiss each other from distance because both controllers transmit the movement information between each other. It's a bit silly, but it reminded me immediately of that.

I think that today, within our limitations, we manage to cover the senses of sight and hearing quite well. We can create incredible effects with sound design and achieve amazing visual effects, but we still can't reach any of the other senses. So, for instance, if I could try on makeup and somehow feel the texture of that gloss on my lips, or perceive some kind of scent — that would be something. Whatever could engage more of the senses to bring greater immersion — I think that's the direction. Maybe what we can develop should aim to create immersion that doesn't rely exclusively on seeing or hearing.

Holly New (VR & Digital Fashion Designer | Founder & CEO of StudioOfNew)

For me, to be augmented is to have the blend of digital and physical. But I don't think the term is correct, it seems to be missing something. I think it would have to be something like mixed touch, but that's also not exact yet. And I know that it kind of means digital touch but it can't be. I think the term is an enigma in my mind, I can precisely figure it out.

But I think being able to feel the digital would be exciting; I just believe it should be implemented in the right way. I think there is a time and a space for touch. There are certain things that you don't need touch for, and there are certain things you do need touch for. And maybe we could have virtual reality without touch and augment reality with touch. And I think it would expand creativity. I mean, the projects I've been doing recently have surpassed anything that I could have ever done physically. And I think

the more that human and technology find its balance, and I think that's the keyword here, balance; creativity will naturally follow. So I think this would open just a whole new world of possibilities.

Swarnim Raj VERMA (3D Modeler Designer • SPORT PRODUCTS DIGITAL CHAIN ASSET DIGIT | Decathlon Lille)

For me, it's kind of like augmented reality, at least in part. It's creating the sensation of touch with all the other senses, almost like simulating a sense by using the others to create a new one.

Frederic FRACHE (Digital Chain Leader • SPORT PRODUCTS DIGITAL CHAIN ASSET DIGIT | Decathlon Lille):

Augmented Touch for me means being able to feel the things we design digitally. It's hard to say what the future is going to look like in 50 years, but I think we are definitely going to see a comeback. Because that's how things work, life is pendular; things we don't think are important today, may become important tomorrow. And I believe we are becoming so saturated with all the screens and information that we are going to see a comeback in the importance of simple things like touching and feeling things around us and being here and now, in the present.

21. (CAUDELL and MIZELL, 1992)

22. TUI - Touch User Interface - is an interface in which digital information and computation are embodied in physical objects and space- so that people can directly touch, manipulate and perceive digital information (MIT MEDIA LAB, [s.d.]).

GLOSSARY: DIGITAL TOOLS

Digital Fabrication — is a process that uses computer-controlled machinery (3D CAD modeling, 3D printing, CNC milling, laser cutting) to create physical objects directly from digital design files (like 3D models or CAD drawings)

AR (Augmented Reality) – An interactive experience that overlays digital elements (images, sounds, textures) onto the real world through devices like smartphones or AR glasses, blending physical and virtual touch sensations.

VR (Virtual Reality) – A fully immersive digital environment that replaces the real world, allowing users to experience and interact with simulated spaces and objects, often enhanced through tactile feedback devices.

AI (Artificial Intelligence) – Systems capable of learning, recognizing patterns, and generating responses or creations. In digital touch contexts, AI can simulate tactile behavior, predict gestures, or design responsive surfaces.

Painter (Substance 3D) – A texturing tool used to apply detailed materials, colors, and tactile effects directly onto 3D models, often mimicking realistic or stylized surface sensations.

Designer (Substance 3D) – A procedural material creation tool for building customizable textures and tactile patterns from scratch, used to define how a surface feels or reacts to touch in digital environments.

Sampler (Substance 3D) – digital application used to quickly and easily create high-quality 3D materials (textures) by sampling, converting, and mixing 2D single photographs into a 3D material.

CLO3D – A 3D garment simulation software that visualizes how fabrics drape, fold, and move, allowing designers to explore the tactile behavior of materials and the sensory experience of clothing in virtual form.

BIBLIOGRAPHY

A23D, [s.d.]. Different Maps in PBR Textures [en ligne]. A23D Blog. Disponible à : https://www.a23d.co/blog/different-maps-in-pbr-textures [consulté le 25 November 2025].

APPLE DEVELOPER, 2025. WWDC25: Meet Liquid Glass [vidéo en ligne]. YouTube. 10 June 2025. Disponible à : https://www.youtube.com/watch?v=lrGYUq1mklk [consulté le 17 October 2025].

BAUDRILLARD, Jean, 1981. Simulacres et Simulation. Paris : Galilée. ISBN 2-7186-0210-4.

BHATTY, Fateh Alam, 2025. My Take on Apple's Liquid Glass [en ligne]. LinkedIn. Disponible à: https://www.linkedin.com/pulse/my-take-apples-liquid-glass-fateh-alam-bhatty-nqdlf [consulté le 12 septembre 2025].

CAMBRIDGE UNIVERSITY PRESS & ASSESSMENT, 2025. Texture [en ligne]. Disponible à : https://dictionary.cambridge.org/us/dictionary/english/texture [consulté le 19 octobre 2025].

CAUDELL, T. P. and MIZELL, D. W., 1992. Augmented reality: an application of heads-up display technology to manual manufacturing processes. In : Proceedings of the Twenty-Fifth Hawaii International Conference on System Sciences, vol. 2, pp. 659–669. Kauai, HI, USA : IEEE. ISBN 978-0-8186-2420-9. DOI 10.1109/HICSS.1992.183317.

CONDILLAC, Étienne Bonnot de, 1798. Traité des sensations, tiré des Œuvres de Condillac, revues, corrigées par l'auteur. Paris : Ch. Houel.

DUNNE, Anthony, 2013. Speculative Everything: Design, Fiction, and Social Dreaming. Cambridge : MIT Press. (The MIT Press Series). ISBN 978-0-262-01984-2.

FERANDO, Christina, 2016. The Deceptive Surface: Perception and Sculpture's "Skin". Images re-vues [en ligne], no. 13. DOI 10.4000/imagesrevues.3931.

FRIESEN, Rebecca Fenton and VARDAR, Yasemin, 2024. Perceived Realism of Virtual

Textures Rendered by a Vibrotactile Wearable Ring Display. IEEE Transactions on Haptics, vol. 17, no. 2, pp. 216–226. DOI 10.1109/TOH.2023.3304899.

FULKERSON, Matthew, 2024. Touch. Stanford Encyclopedia of Philosophy [en ligne]. First published 25 November 2015; substantive revision 12 June 2024. Disponible à : https://plato.stanford.edu/entries/touch/ [consulté le 16 septembre 2025].

GIBSON, James J., 1986. The Ecological Approach to Visual Perception. New York : Psychology Press. ISBN 978-89859-959-6.

HANSEN, Mark B. N., 2012. Bodies in Code: Interfaces with Digital Media. London : Routledge. ISBN 978-0-203-94239-0.

HOST HAVOC, 2025. Minecraft User and Revenue Statistics (Updated 2025) [en ligne]. Host Havoc. [Posted 4 September 2025]. Disponible à : https://hosthavoc.com/blog/minecraft-statistics [consulté le 24 November 2025].

IKEA, [s.d.]. Bring your space to life with IKEA Kreativ [en ligne]. Ikea.com. Disponible à : https://www.ikea.com/au/en/campaigns/bring-your-space-to-life-with-ikea-kreativ-pub3fa91c50/ [consulté le 22 septembre 2025].

KODAK, Bence L. and VARDAR, Yasemin, 2023. FeelPen: A Haptic Stylus Displaying Multimodal Texture Feels on Touchscreens. IEEE/ASME Transactions on Mechatronics, vol. 28, no. 5, pp. 2930–2940. DOI 10.1109/TMECH.2023.3264787.

KUSWAHYUNI, Syarafina, 2025. Liquid Glass isn't a design failure—it is Apple's most tactile digital future yet [en ligne]. UX Collective. [Posted 27 October 2025].

LÉVY, Pierre, 1998. Becoming Virtual: Reality in the Digital Age. New York : Plenum Trade. ISBN 978-0-306-45788-3.

MARKS, Laura U., 2002. Touch: Sensuous Theory and Multisensory Media. Minneapolis : University of Minnesota Press. ISBN 978-0-8166-3888-8.

MERLEAU-PONTY, Maurice, 1968. The Visible and the Invisible. Evanston (IL) : Northwestern University Press. ISBN 978-0-8101-0454-0.

MERLEAU-PONTY, Maurice and SMITH, Colin, 2006. Phenomenology of Perception: An Introduction. Repr. London : Routledge. (Routledge Classics). ISBN 978-0-415-27840-9. (Note: The two Merleau-Ponty entries are correctly sorted chronologically.)

MINECRAFT WIKI, 2025. Minecraft Wiki [en ligne]. Disponible à: https://minecraft.fandom.com/wiki/Minecraft_Wiki [consulté le 15 septembre 2025].

MIT MEDIA LAB, 2025. Research – Groups & Centers. MIT Media Lab [en ligne]. Disponible à : https://www.media.mit.edu/research/?filter=groups_centers [consulté le 24 octobre 2025].

MIT MEDIA LAB, [s.d.]. Materiable - Tangible Media Group, MIT Media Lab. Disponible à : https://tangible.media.mit.edu/project/materiable/ [consulté le 25 novembre 2025].

MOJANG AB, [s.d.]. About Minecraft [en ligne]. Minecraft.net. Disponible à : https://www.minecraft.net/fr-fr/about-minecraft [consulté le 25 novembre 2025].

NAKAGAKI, Ken et al., 2016. Materiable: Rendering Dynamic Material Properties in Response to Direct Physical Touch with Shape-Changing Interfaces. In : Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems, pp. 2764–2772. San Jose, California, USA : ACM. ISBN 978-1-4503-3362-7. DOI 10.1145/2858036.2858104.

NOVA, Nicolas et al., 2012. Curious Rituals: Gestural Interaction in the Digital Everyday [en ligne]. Pasadena (CA): Art Center College of Design; Hear Future Laboratory. Disponible à : http://curiousrituals.wordpress.com/ [consulté le 25 November 2025].

OCTANE RENDER, OTOY, Inc., [s.d.]. [Image: Octane Renderer PBR Material Inputs] [en ligne]. OTOY Help Center. Disponible à : https://help.otoy.com/hc/article_attachments/22286375135771 [consulté le 25 November 2025].

PINK, Sarah, 2015. Doing Sensory Ethnography. 2nd ed. London : SAGE Publications Ltd. ISBN 978-1-4462-8759-0.

REDDIT, 2022. How much textures are there in minecraft [en ligne]. Reddit. [Posted 26 May 2022]. Disponible à : https://www.reddit.com/r/Minecraft/comments/uy6ixt/how_much_textures_are_there_in_minecraft/ [consulté le 24 November 2025].

RYAN, Marie-Laure, 2010. Narrative as Virtual Reality: Immersion and Interactivity in Literature and Electronic Media. Repr. Baltimore (Md.) : Johns Hopkins University Press. (Parallax). ISBN 978-0-8018-6487-2.

TABLE OF FIGURES

Figure 1. 3D print over fabric. Source: Personal work.
0005

Figure 2. Motor habits linked to touchscreen gestures. Source: (NOVA et al., 2012).
0015

Figure 3. Rough to smooth. Source: Source: (A23D, s.d.).
0018

Figure 4. 3D 2. Digital material and matter. Source: Personal work.
0020

Figure 5. Texture layers. Source: Source: (OCTANE RENDER, OTOY, Inc., s.d.).
0022

Figure 6. Minecraft world. Source: (MOJANG AB, s.d).
0025

Figure 7. Minecraft grass blocks. Source: (MINECRAFT WIKI, s.d.).
0028

Figure 8. Liquid glass button. Source: (APPLE DEVELOPER, 2025).
0031

Figure 9. Liquid glass icon. Source: (KUSWAHYUNI, 2025).
0034

Figure 10. Screenshot of IKEA Kreativ. Source: (IKEA, 2025)
0036

Figure 11. Umbra. Source: Personal work.
0044

Figure 12. Leman Pulse. Source: Personal work.
0046

Figure 13. L.O.U. Source: Personal work.
0048

Figure 14. Materiable. Source: (MIT MEDIA LAB, s.d)
0050